



Synergy-incorporated Bayesian Petri Net: A method for mining “AND/OR” relation and synergy effect with application in probabilistic reasoning

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ABSTRACT

Bayesian networks (BNs) are widely used for knowledge representation and reasoning. However, they suffer from the following limitations: 1) They are unable to explicitly learn “AND” relations and synergy effects from data; 2) They do not depict “AND” relations among causes directly, which sometimes leads to the high complexity of constructing conditional probability tables (CPTs); and 3) During their knowledge representation and reasoning, they fail to express any synergy (catalytic or inhibitory) effect that a non-causal variable may have on a reasoning rule. To address the mentioned issues, this study proposes a method for mining and modelling “AND/OR” relations and synergy effects called Synergy-incorporated Bayesian Petri Net (SBPN). The method integrates a BN with Petri net concepts. It can directly learn “AND/OR” relations and synergy effects from data, harnessing the power of Petri nets to model “AND/OR” relations in BNs, resulting in a compact CPT. Synergy effects are easily depicted via the SBPN, and their influence on the reasoning procedure can be formally expressed using the SBPN. Experimental results on open datasets demonstrate the effectiveness and advantages of the proposed method for knowledge mining and probabilistic reasoning.

1. Introduction

Knowledge mining, representation, and reasoning constitute the core of knowledge engineering. Knowledge mining aims to extract valuable knowledge from data [1–3]. Knowledge representation encodes the acquired knowledge in a structured manner [4–6]. Knowledge reasoning addresses domain-specific problems through a specific reasoning mechanism [7–9].

Various knowledge representations and reasoning methods have been proposed for different decision-making scenarios, including Bayesian networks (BNs) [10,11,14,41], neural networks [42], fault trees [43], fuzzy rules [44], and Petri nets (PNs) [32,33]. Among them, the BN is a powerful probabilistic graphical model that is well suited for reasoning under uncertainty. It consists of a directed acyclic graph that depicts the dependency relationships among variables and a set of conditional probability distributions that quantify the strength of these dependencies. Conditional and joint probability distributions are used to implement probabilistic reasoning. For any variables v_i and v_j in the BN, if there is a directed arc from v_i to v_j , v_i is identified as the parent (i.e., cause) of v_j . However, the BN

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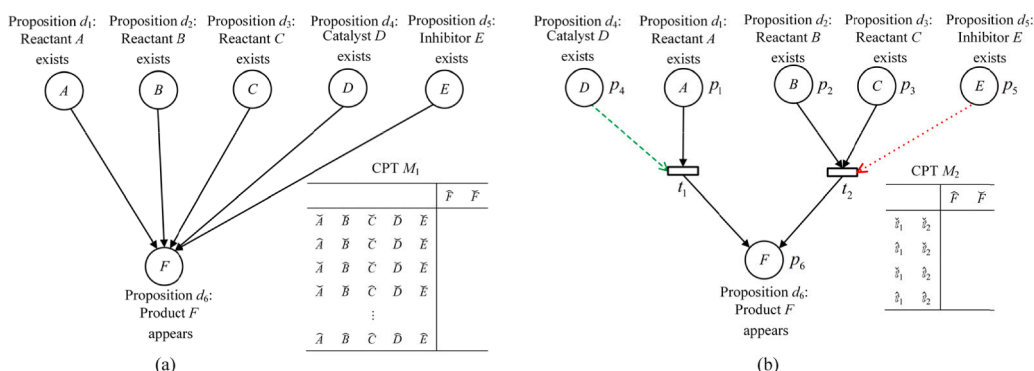


Fig. 1. Difference between a traditional BN and the proposed modelling paradigm. (a) BN model instance and (b) proposed modelling paradigm.

does not directly depict the “AND” relations among causes or express any catalytic or inhibitory effect that a non-causal variable may have on a reasoning rule. Consider the following two chemical reactions: 1) Reaction $a: A \xrightarrow{D} F$, where A , D , and F are the reactant, catalyst (a substance that increases the rate of a chemical reaction without itself being consumed), and product of a , respectively; and 2) Reaction $b: B + C \xrightarrow{E} F$, where B and C are the reactants of b , and E and F are its inhibitor (a substance that slows down a chemical reaction or prevents it from occurring) and product, respectively. Reactants A , B , and C are the determining factors influencing the production of F , whereas catalyst D and inhibitor E play a secondary and non-decisive role in its production. Essentially, A , B , and C are the causes of the production of F , whereas D and E are not. However, D and E can affect the production of F . Specifically, under the action of catalyst D , reaction a is facilitated, meaning D has a catalytic effect on a . On the contrary, reaction b is more difficult to carry out in the presence of inhibitor E , indicating that E has an inhibitory effect on b . In a rule-based system, such phenomena can be expressed as follows: A non-causal variable can affect the probability that a reasoning rule holds. In this study, if a non-causal variable can increase (reduce) the probability that a reasoning rule holds, the variable is said to have a catalytic (inhibitory) effect on the rule. Both catalytic and inhibitory effects are referred to as synergy effects in this study.

The causal relations among A , B , C , and F can be expressed by the following causal rules: R_1 : IF “Reactant A exists” THEN “Product F appears” and R_2 : IF “Reactant B exists” AND “Reactant C exists” THEN “Product F appears”. In addition, proposition “Catalyst D exists” has a catalytic effect on R_1 , meaning it can increase the probability that R_1 holds. Moreover, proposition “Inhibitor E exists” has an inhibitory effect on R_2 , i.e., it can reduce the probability that R_2 holds.

Although the rules mentioned above can express “AND” relations, they fail to characterize synergy effects and perform probabilistic reasoning for such relations and effects. BNs are graphical models that integrate probabilistic reasoning. However, they are unable to directly depict the above knowledge. One possible way to model the above knowledge using a BN is shown in Fig. 1(a). Throughout this paper, $\bar{s}$ or $\bar{s}$ denotes the state of a variable, place or transition corresponding to s . For a variable or place, its state is “true” or “false”. As for a transition, its state is “enabled” or “disabled”. From Fig. 1(a), the BN fails to directly depict the “AND” relation between variables B and C . Additionally, it does not model the catalytic effect of variable D and the inhibitory effect of variable E in a straightforward way. This leads to high complexity in constructing the conditional probability table (CPT) M_1 . In view of this issue, this study aims to incorporate the concept of PN [12] into BNs. A PN comprises three types of elements: place, transition, and flow relation. When modelling a rule-based system using a PN, a place represents a proposition variable, a transition represents an indecomposable reasoning rule, and the flow relations from place to transition and from transition to place define the causal direction between the propositions. Propositions that point to the same transition have an “AND” structure. By integrating all causal variables in an “AND” structure into a single variable represented by a PN transition, the size of the BN’s CPT can be reduced. However, traditional PNs fail to consider synergy effects in knowledge representation and reasoning. To address the above issues, this study proposes a new modelling method that considers “AND/OR” relations as well as synergy effects by fusing a PN and BN, as illustrated in Fig. 1(b).

In Fig. 1(b), the PN place (represented by a circle) is used to depict the causal variable, effect variable, or synergy factor. A PN transition (represented by a rectangle) is used to model the causal rule. The causal variable is expressed as the input place of a transition. Causal variables that point to the same transition have an “AND” relation. The effect variable is expressed as the output place of a transition. A solid arc represents the flow relation from a causal variable to a transition (or from a transition to an effect variable). Furthermore, the flow relation from a synergy factor to a transition is represented by a dashed or dotted arc. The green dashed arc indicates the catalytic effect, and the red dotted arc represents the inhibitory effect. Different types of flow relations play different roles in knowledge reasoning. As shown in Fig. 1(b), transitions t_1 and t_2 are used to model R_1 : IF “Reactant A exists” THEN “Product F appears” and R_2 : IF “Reactant B exists” AND “Reactant C exists” THEN “Product F appears”, respectively. Places p_4 and p_5 are used to model the synergy factors of R_1 and R_2 , respectively. Other places model the cause and effect variables. The probability that proposition “Product F appears” holds depends on the probability that either R_1 or R_2 is activated. Therefore, the CPT M_2 can be determined by identifying whether R_1 or R_2 is activated. Clearly, the complexity of constructing M_2 is lower than that of constructing M_1 . A leaky noisy-OR model [15] is utilized as the reasoning mechanism in this study. Hence, CPT M_2 in Fig. 1(b) can be simplified further, resulting in CPT $m(p_6)$, as illustrated in Fig. 2(d).

Next, we discuss why mining and modelling “AND/OR” relations and synergy effects is an important issue. The importance of

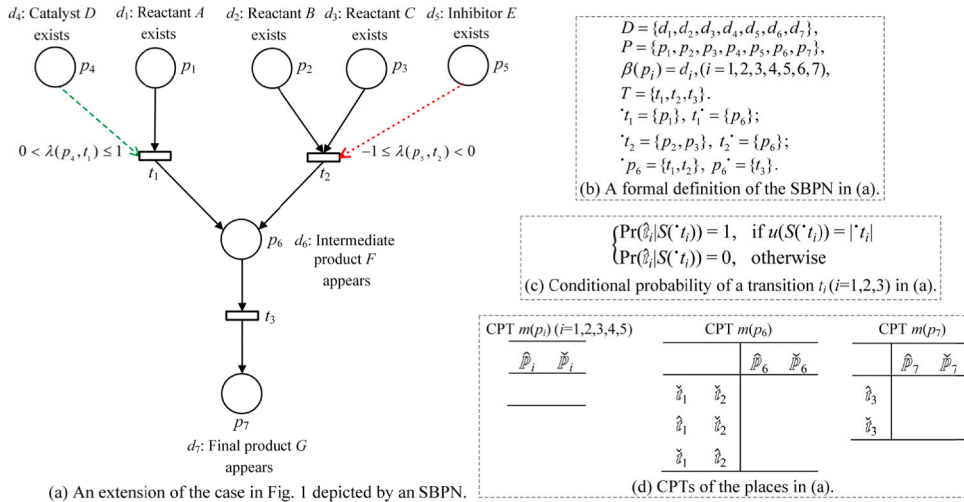


Fig. 2. SBPN-based knowledge representation for an extension of the case in Fig. 1.

mining and modelling “AND/OR” relations is illustrated first. As shown in Fig. 1, the complexity of constructing a CPT can be reduced by integrating the causal variables within the same “AND” structure into a single variable. In addition, “AND/OR” relation is a complex causal relation that widely exists in the real world. By distinguishing between “AND” relations and “OR” relations in knowledge representation, one can intuitively understand the causal mechanisms. Consider for instance Fig. 1(b). There is an “AND” relation between B and C , which means that F exists if and only if B and C exist simultaneously. That is, when B exists but C does not, F cannot be generated. However, it is difficult to depict such a causal mechanism intuitively using a BN graph structure. Next, the necessity of mining and modelling the synergy effects is illustrated. Suppose that all substances A – F depicted in Fig. 1 exist. Since BNs fail to consider the synergy effect of a non-causal variable on a reasoning rule, based on the causal structure in Fig. 1(a), A – E are all factors contributing to the generation of F . However, according to the causal structure in Fig. 1(b), the presence of E can greatly reduce the probability that rule R_2 : IF “Reactant B exists” AND “Reactant C exists” THEN “Product F appears” holds, whereas the presence of D can significantly increase the probability that rule R_1 : IF “Reactant A exists” THEN “Product F appears” holds. In this case, A is the reason for the generation of F . Therefore, mining and modelling “AND/OR” relations and synergy effects can help one better understand the causal mechanisms.

Here, we discuss another important issue: What do PNs offer that BNs cannot model? From the perspective of knowledge representation, one can use BN to model “AND” relations by integrating all causal variables within the same “AND” structure into an additional BN variable. However, from the perspective of knowledge mining, BN structure learning methods [24,25,45] focus on learning the causal relations between observed variables. Because the additional variable does not exist in the original data, it is impossible to directly determine the causal relations between it and the other variables in the BN from the data. Modelling “AND” relations using PN transitions does not have this limitation, as they are employed to constrain the causal relations between observed variables. In fact, one can also implicitly express “AND/OR” relations in the form of functional BNs [48], which can effectively discover causal relations based on appropriately hypothetical causal functions. However, it does not focus on learning “AND” relations and synergy effects from the data, which is the focus of this study. To present “AND/OR” relations and synergy effects in an understandable and intuitive way, this study explicitly depicts them as graph structures. Specifically, PN transitions are employed as additional nodes in the BN for modelling “AND/OR” relations as well as synergy effects.

The above statements introduce the limitations of BN in modelling “AND/OR” relations and synergy effects. Sometimes, they cannot be obtained from prior knowledge and must be learned from the data. However, existing causal discovery methods, such as constraint-based BN structure learning [24–27], score-based BN structure learning [45–47], and functional causal models [28–30], focus on mining causal relations but fail to consider “AND/OR” relations and synergy effects. Related to this limitation, some research efforts have aimed to discover the combined effects from the data [16,17]. The so-called combined effects are similar to the “AND” relations discussed in this paper. They focus on mining the “AND” relation among non-causal variables. That is, each of these variables individually cannot be identified as a causal variable, while their combination can be detected as a causal variable. However, for multiple variables that form an “AND” relation toward a target variable, each of them may be individually identified as a causal variable of the target variable.¹ In such situations, the “AND” relation among them cannot be identified by the mining methods in [16,17]. In this study, we focus on how to identify the “AND” relation among such variables. In addition, the work [17] mines the enhancement effects among the causal and non-causal variables, which is similar to the catalytic effect defined in this study. However, in some practical situations, such as the chemical reactions mentioned above, there may be an enhancement effect of a non-causal

¹ We have verified that such variables may show strong correlations with the target variable in data. Thus, traditional causal BN learning methods [15,24] are likely to identify each of them as a causal variable of the target one.

variable on a reasoning rule rather than on a variable, which is difficult to capture using the mining method in [17].

Knowledge mining and modelling play crucial roles in knowledge engineering. Using a constructed knowledge model to implement probabilistic reasoning is important. However, existing BN reasoning methods [38,39,41] fail to consider the synergy effect of a non-causal variable on a reasoning rule, which affects the credibility of probabilistic reasoning. Related to this limitation, some BN-related studies explored the impact of enhancement or weakening effects on probabilistic reasoning among causal variables [18–20]. However, a non-causal variable may deeply affect reasoning results through a catalytic or inhibitory effect. Existing BN reasoning methods neglect these synergy effects in their reasoning procedures.

“AND/OR” relations discussed in this paper have attracted the interest of researchers in the field of PN. A fuzzy Petri net (FPN) [21,22], which is an extension of the PN, has been proposed for fuzzy knowledge representation and reasoning. It can portray fuzzy rules containing “AND/OR” relations in a precise and structured manner, and has an efficient fuzzy reasoning mechanism. The work [23] presents an extended FPN that incorporates synergy effects into FPN-based knowledge representation and reasoning. However, existing FPN works only discuss “AND/OR” relations and synergy effects in knowledge representation and reasoning. They do not involve how to identify “AND/OR” relations or synergy effects from data. In fact, the modelling of “AND/OR” relations and synergy effects in [21–23] relies heavily on expert experience. In contrast, our proposed method can directly learn “AND/OR” relations and synergy effects from data. In addition, the reasoning parameters in [21–23] are subjectively set by experts, whereas those in this study can be objectively obtained from the data. Clearly, our method is more reliable than those of [21–23].

The concept of combining PN and BN has been discussed in [38,39]. The work [38] first constructed a fault tree for a pump failure scenario based on domain knowledge and converted the tree into a stochastic Petri net (SPN). The Bayes theory was then introduced into the SPN simulation analysis to enhance the modelling ability of the SPN for pump failure analysis. In [39], PN and BN structures were constructed for the risk analysis of fire and explosion scenarios based on domain knowledge. Specifically, the PN was adopted to calculate the probability of accidents, whereas the BN was utilized to determine the domino effect probabilities. Although these two studies involve the idea of combining PN and BN, they cannot automatically construct a causal network capable of handling “AND/OR” structures and synergy effects from data. In essence, they employed a BN as a probabilistic reasoning tool for PN simulation analysis and followed the probability distribution and reasoning mechanism of a traditional BN. In contrast, this study redefines the probability distribution and reasoning mechanism of BN via PN transition, which can effectively capture the impact of synergy effects on “AND/OR” relations during probabilistic reasoning. In essence, this study focuses on how to mine “AND/OR” relations and synergy effects from the data, as well as how these synergy effects affect probabilistic reasoning. This is fundamentally different from the focus and purpose of [38,39].

In summary, to effectively characterize the “AND/OR” relations among causes and synergy effects in knowledge mining, representation, and reasoning, this study proposes a method for mining and modelling “AND/OR” relations and synergy effects called Synergy-incorporated Bayesian Petri Net (SBPN). This study makes the following new contributions to the field of knowledge engineering.

- 1) It learns complex “AND/OR” relations that exist in common causal relations,² as well as the synergy effects imposed on “AND/OR” relations from the data.
- 2) It incorporates the idea of PN transition into the probability distribution of BN and accurately models “AND/OR” relations and synergy effects, resulting in a compact CPT.
- 3) It quantifies the effect of synergy effects on probabilistic reasoning, and its reasoning parameters are learned from the data.

The rest of this paper is organized as follows. Section 2 reviews the related work on BN modelling and reasoning, PN modelling, and causal discovery. Section 3 presents the formal definition of SBPN and its representation paradigm for “AND/OR” relations and synergy effects. Section 4 presents the mining ideas for “AND/OR” relations and synergy effects. Section 5 introduces probabilistic reasoning that considers synergy effects. Section 6 evaluates the performance of the SBPN in knowledge mining and probabilistic reasoning. Finally, Section 7 concludes the study.

2. Preliminaries and related work

This research involves BN modelling and reasoning, PN modelling, and causal discovery. They are discussed in the following sections.

2.1. Bayesian network modelling and reasoning

As a probability graph model, the BN [11] encodes the joint probabilistic relationships among random variables. Formally, it can be defined as $B = (G, Q)$, where $G = (V, E)$ is a directed acyclic graph and $Q = \bigcup_{i=1}^n \{q_i\}$ is a set of conditional probability distributions. In particular, $V = \bigcup_{i=1}^n \{v_i\}$ denotes a set of random variables (each variable corresponds to a node in the graph). $E = (e_{ij}) \rightarrow \{0, 1\}$ defines the dependence relationship between the variables v_i and v_j (if $e_{ij} = 0$, v_i and v_j are independent; otherwise, v_i is identified as a parent of v_j). $q_i (i = 1, 2, \dots, n)$ encodes the conditional probability of v_i given its parents (denoted by $P_A(v_i)$). If v_i has no parent, q_i indicates its

² In this paper, a common causal relation refers to a causal relation without considering “AND/OR” relations among causes and synergy effects.

prior probability. Let $\Pr$ denote a probability distribution. The joint probability distribution (JPD) over variable set V is defined as

$$\Pr(v_1, v_2, \dots, v_n) = \prod_{i=1}^n \Pr(v_i | P_A(v_i)). \quad (1)$$

The complexity of constructing the CPT for the variable v_i increases exponentially with the number of $P_A(v_i)$'s. A noisy-OR model [13] can alleviate this burden by assuming a disjunctive interaction among all causal variables of the effect variable. Its reasoning idea can be described as

$$\Pr(\hat{e} | S(C_e)) = 1 - \prod_{j \in J} (1 - \Pr(\hat{e} | \check{c}_1, \dots, \check{c}_{j-1}, \hat{c}_j, \check{c}_{j+1}, \dots, \check{c}_n)) \quad (2)$$

where $\hat{e}$ represents the presence of an effect variable e , $S(C_e)$ indicates the states of all e 's causal variables $C_e = \{c_1, c_2, \dots, c_n\}$ (state $\check{c}_i$ indicates the absence of cause c_i ; state $\hat{c}_j$ indicates the presence of cause c_j), and J denotes the indices of the causes that are present.

A noisy-OR model assumes that an effect e cannot occur if all of its known causes C_e are absent, i.e., $\Pr(\hat{e} | \check{c}_1, \dots, \check{c}_n) = 0$. Generally, due to the existence of unobserved variables, $\Pr(\hat{e} | \check{c}_1, \dots, \check{c}_n) \neq 0$. The leaky noisy-OR model [15], employed as the reasoning framework in this study, considers the impact of unobserved variables on probabilistic reasoning. Its reasoning mechanism can be expressed as

$$\Pr(\hat{e} | S(C_e)) = 1 - (1 - \Pr(\hat{e} | \check{c}_1, \dots, \check{c}_n)) \prod_{j \in J} (1 - \Pr(\hat{e} | \check{c}_1, \dots, \check{c}_{j-1}, \hat{c}_j, \check{c}_{j+1}, \dots, \check{c}_n)). \quad (3)$$

There may be enhancement or weakening effects among causes. Consider an effect variable e with two causes c_i and c_j , if $\Pr(\hat{e} | \hat{c}_i, \hat{c}_j) > \Pr(\hat{e} | \hat{c}_i)$ and $\Pr(\hat{e} | \hat{c}_i, \hat{c}_j) > \Pr(\hat{e} | \hat{c}_j)$, there is an enhancement effect between c_i and c_j . If $\Pr(\hat{e} | \hat{c}_i, \hat{c}_j) < \Pr(\hat{e} | \hat{c}_i)$ and $\Pr(\hat{e} | \hat{c}_i, \hat{c}_j) < \Pr(\hat{e} | \hat{c}_j)$, then a weakening effect exists between them. The noisy-OR and leaky noisy-OR models can depict the enhancement effects among causes, but fail to express the weakening effects among causes. Some studies [18–20] consider the weakening effects among causes. However, as mentioned in Section 1, they neglected the impact of a non-causal variable on a reasoning rule during probabilistic reasoning. To address this issue, a probabilistic reasoning framework that considers these impacts is proposed in this study.

This study introduces PN transition into BN to model “AND” relations. Next, the concept of a PN is introduced.

2.2. Petri net modelling

For propositions a and b , if a can lead to b , their relationship can be expressed as rule R : IF a THEN b , where a is the antecedent proposition of R and b is the consequent proposition of R . In fact, R can be modelled by a PN transition t . Thus, a and b are called the antecedent and consequent propositions of t , respectively. Next, a brief introduction to the PN is provided.

PN [12,32,33] is a modelling paradigm that combines a rich mathematical analysis tool with graphical representation. It is widely used to model and analyze systems containing sequential, concurrent, and conflicting relations, which characterize complex logical relations and should be considered in knowledge mining and reasoning. Formally, a PN can be defined as a 4-tuple: $PN = (P, T, F, M_0)$ where

- 1) $P = \{p_1, p_2, \dots, p_m\}$ is a finite set of places.
- 2) $T = \{t_1, t_2, \dots, t_n\}$ is a finite set of transitions, which satisfies $P \cup T \neq \emptyset$ and $P \cap T = \emptyset$.
- 3) $F \subseteq (P \times T) \cup (T \times P)$ is a set of flow relations, which defines arcs from places to transitions or from transitions to places.
- 4) $M_0 : P \rightarrow \{0, 1\}$ is the initial marking of a PN, which describes the status of the net system.

This study makes the following adjustments to the above definition. Each place is used to represent a proposition. Each transition is used to represent a reasoning rule. If the proposition corresponding to place p is the antecedent proposition of the rule corresponding to transition t , $(p, t) \in F$. If the proposition corresponding to p is the consequent proposition of the rule corresponding to t , $(t, p) \in F$. If a reasoning rule has multiple antecedent propositions, there is an “AND” relation among them. In this case, the transition corresponding to the rule has multiple input places. If and only if all the antecedent propositions of a transition hold, then the transition is enabled, meaning that the rule corresponding to the transition is activated. The marking of a place encodes its conditional probability distribution, which is formally defined in Section 3.1.

2.3. Causal discovery

To model “AND/OR” relations and synergy effects, sometimes one needs to discover them from the data. Because they are defined based on causal relations, causal relations should first be discovered from the data. The related studies on causal discovery are discussed next.

Causal discovery, which is a popular research topic in knowledge mining, plays a key role in the development of artificial intelligence. It aims to mine the causal relations among random variables from the observed data. Many causality mining methods have been proposed, such as constraint-based BN structure learning [24–27], score-based BN structure learning [45–47], and functional

causal models [28–30]. Constraint-based BN structure learning, such as the PC algorithm [24,25], utilizes a series of conditional hypothesis tests and constraint rules to determine the causal relations among variables. Score-based BN structure learning aims to discover causal structures by optimizing a properly defined score function. Among them, greedy equivalence search (GES) [45], globally optimal Bayesian network learning using integer linear programming (GOBNILP) [46], and hill climbing (HC) [47] are three important BN structure learning approaches. The GES directly performs a search over the space of equivalence classes. The GOBNILP uses an integer linear programming formulation to learn the optimal BN structures. The HC implements a greedy local search by considering only local changes (i.e., neighbor networks). Functional causal models focus on identifying causal directions by exploiting the asymmetry of causality. Among them, the additive noise model (ANM) [28,29] is a classical causal discovery method that utilizes the asymmetry of noise in a causal relation to determine the causal direction. For a detailed introduction to classical causal discovery methods and their underlying causal assumptions, please refer to the survey work [49].

As shown in Fig. 1(b), there may be “AND/OR” relations among causal variables and catalytic or inhibitory effects imposed on rules. However, existing causal discovery methods are unable to capture such “AND/OR” relations and synergy effects. Related to this limitation, based on the HITON-PC algorithm [31] (a local causal discovery method based on the PC algorithm), the work [16] proposes a multi-level HITON-PC (MHPC) method to discover the “AND” relations among non-causal variables. The work [17] extends the ANM to discover the “AND” relations among non-causal variables as well as the enhancement effects among causal and non-causal variables. However, as mentioned in Section 1, they fail to depict the “AND” relations among causal variables and the catalytic or inhibitory effects imposed on rules. To address this issue, this study focuses on capturing such “AND” relations and synergy effects from the data.

3. Synergy-incorporated Bayesian Petri net (SBPN)

Traditional probabilistic graphical models do not directly model “AND” relations, which makes the size of a CPT grow exponentially. They also fail to consider the impact of synergy effects on probabilistic reasoning, especially when a synergy factor is associated with an “AND/OR” relation. To address these two issues, this study proposes a method for mining and modelling “AND/OR” relations and synergy effects called SBPN. The basic idea of this study is as follows. First, existing causal discovery algorithms [24,28] are used to learn common causal relations from the data, resulting in a BN. Then, the “AND/OR” relations among the causes of BN as well as the synergy effects imposed on “AND/OR” relations are discovered from the data, and PN transitions are employed to model such relations and effects in BN. Then, by integrating all the causal variables in an “AND” structure into a single variable, the size of BN’s CPT is reduced. Based on this, PN transitions are introduced into the reasoning mechanism of the BN. This enables BN to characterize the impact of synergy effects on “AND/OR” relations during probabilistic reasoning. Next, a formal definition of the SBPN and its causality representation paradigm are presented.

3.1. Definition of SBPN

Definition 1 (SBPN). An SBPN is an 8-tuple: $SBPN = (D, P, \beta, T, F, Pr, \lambda, M)$ where

- 1) $D = \{d_1, d_2, \dots, d_m\}$ is a finite set of propositions.
- 2) $P = \{p_1, p_2, \dots, p_m\}$ is a finite set of places. There is a one-to-one mapping between P and D .
- 3) $\beta : P \rightarrow D$ is an association function, which defines a bijective mapping from places to propositions. Each place in P corresponds to a proposition in D .
- 4) $T = \{t_1, t_2, \dots, t_n\}$ denotes a finite set of transitions, which satisfies $P \cap T \cap D = \emptyset$. Each transition represents a reasoning rule.
- 5) $F \subseteq (P \times T) \cup (T \times P)$ is a set of flow relations, which defines arcs from places to transitions or from transitions to places. The pre-set of a node $x \in P \cup T$ is defined as $\bullet x = \{y \in P \cup T | (y, x) \in F\}$. The post-set of a node $x \in P \cup T$ is defined as $x^\bullet = \{y \in P \cup T | (x, y) \in F\}$. If the proposition corresponding to a place p , i.e., $\beta(p)$, is the antecedent proposition of the rule corresponding to a transition t , then $p \in \bullet t$ and $t \in p^\bullet$ hold; if proposition $\beta(p)$ is the consequent proposition of the rule corresponding to t , then $p \in t^\bullet$ and $t \in \bullet p$ hold.
- 6) $Pr : P \cup T \rightarrow [0, 1]$ is a probability function that maps each place and transition to a probability value in $[0, 1]$. In this study, $Pr(\hat{p})$ represents the probability that $\beta(p)$ is true, and $Pr(\hat{t})$ indicates the probability that t is enabled (a transition is enabled if all its input place-associated propositions, that is, its antecedent propositions, hold simultaneously). The probability of a transition being enabled is equivalent to the probability that all its antecedent propositions hold simultaneously.
- 7) $\lambda : P \times T \rightarrow [-1, 1]$ is a function that portrays the strength of a synergy effect. Concretely, $\lambda(p, t)$ is called a synergy coefficient that indicates the strength of the catalytic or inhibitory effect of $\beta(p)$ on t . $\lambda(p, t)$ is greater (or less) than 0 iff $\beta(p)$ has a catalytic (or inhibitory) effect on t . Both catalytic and inhibitory effects are called synergy effects. The stronger the synergy effect, the greater the absolute value of the synergy coefficient.
- 8) $M = \bigcup_{i=1}^m \{m(p_i)\}$ is the marking of an SBPN, where $m(p_i)$ encodes the conditional probability distribution of a place p_i given its input transitions. If p_i has no input transition, $m(p_i)$ encodes its prior probability.

Note that: 1) An SBPN should be a directed acyclic graph; 2) In this paper, $\hat{p}$ (or $\bar{p}$) indicates that the proposition $\beta(p)$ is true (or false). $\hat{t}$ (or $\bar{t}$) indicates that the transition t is (or is not) enabled; and 3) An enabling transition activates its corresponding reasoning rule. This is a necessary prerequisite for a synergy effect to work.

This study focuses on mining and modelling “AND/OR” relations as well as synergy effects for binary variables (i.e., the variables that only take two values: 0 and 1). The noisy-OR model [13], which is a classic BN modelling and reasoning paradigm for binary variables, is employed as the modelling and reasoning mechanism in this study. In particular, PN transitions are added to the noisy-OR model as additional nodes, which can further reduce CPT size. The conditional probability of the transition being enabled is defined based on the noisy-OR model. In addition to the noisy-OR model, various types of BNs have been developed to solve domain-specific problems [38,39,41]. For instance, a hybrid BN [41] was proposed to model the causal relations among discrete and continuous variables and perform inference for such variables. It is a powerful probabilistic graphical model widely used to model complex systems. However, it is unable to discover “AND/OR” relations and synergy effects from the data. When modelling causal relations using it, one cannot directly distinguish between “AND” relations and “OR” relations by using the functional relationships in it. In addition, it fails to capture the synergy effect of a non-causal variable on a reasoning rule during its reasoning procedure. Different from it, by introducing PN transitions into the reasoning mechanism of the noisy-OR model, the SBPN can accurately depict such effect during probabilistic reasoning.

Mathematically, the JPD of SBPN is defined as

$$\Pr(x_1, x_2, \dots, x_r) = \prod_{i=1}^r \Pr(x_i | \bullet x_i) \quad (4)$$

where x_i represents a place or transition, $\bullet x_i$ indicates the input places of a transition or the input transitions of a place, and r is the total number of places and transitions. To compute a JPD, the conditional probability of each transition given its input places, and that of each place given its input transitions, should be determined first. The ideas for calculating these two types of conditional probabilities are introduced below.

First, the probability of enabling a transition is computed as follows. For a transition t_i , let $S(\bullet t_i) = \bigcup_{p_i \in \bullet t_i} \{s(p_i)\}$ denote the states of all its input places, where $s(p_i)$ represents the state of place p_i and satisfies $s(p_i) = \hat{p}_i$, $\bar{p}_i$, where $\hat{p}_i$ (or $\bar{p}_i$) indicates that proposition $\beta(p_i)$ holds (or does not hold). Next, let $u(S(\bullet t_i))$ be a statistical function that represents the number of t_i 's antecedent propositions that hold at state $S(\bullet t_i)$. The probability that t_i is enabled at $S(\bullet t_i)$ is defined as

$$\begin{cases} \Pr(\hat{t}_i | S(\bullet t_i)) = 1, & \text{if } u(S(\bullet t_i)) = |\bullet t_i| \\ \Pr(\hat{t}_i | S(\bullet t_i)) = 0, & \text{otherwise} \end{cases} \quad (5)$$

where $|\bullet t_i|$ represents the total number of t_i 's antecedent propositions. From (5), if $u(S(\bullet t_i)) = |\bullet t_i|$ (i.e., all the antecedent propositions of t_i hold simultaneously), $\Pr(\hat{t}_i | S(\bullet t_i)) = 1$ (i.e., t_i is enabled at $S(\bullet t_i)$). Otherwise, $\Pr(\hat{t}_i | S(\bullet t_i)) = 0$, i.e., t_i cannot be enabled at $S(\bullet t_i)$.

An extension of the case discussed in Fig. 1 is shown in Fig. 2, which is used to illustrate how to construct the CPT of a place. Consider p_6 in Fig. 2(a). Its CPT, denoted by $m(p_6)$, is shown in Fig. 2(d). Different from the traditional CPT, $m(p_6)$ is defined on transitions t_1 and t_2 . In fact, $\Pr(\hat{t}_1) = \Pr(\hat{p}_1)$, i.e., t_1 is enabled as long as proposition $\beta(p_1)$ holds; $\Pr(\hat{t}_2) = \Pr(\hat{p}_2, \hat{p}_3)$, i.e., t_2 is enabled iff both propositions $\beta(p_2)$ and $\beta(p_3)$ hold simultaneously. Consequently, $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2) = \Pr(\hat{p}_6 | \hat{p}_1, \hat{p}_2, \hat{p}_3) + \Pr(\hat{p}_6 | \hat{p}_1, \hat{p}_2, \bar{p}_3) + \Pr(\hat{p}_6 | \bar{p}_1, \hat{p}_2, \hat{p}_3) + \Pr(\hat{p}_6 | \bar{p}_1, \hat{p}_2, \bar{p}_3)$, and $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2) = \Pr(\hat{p}_6 | \hat{p}_1, \hat{p}_2, \hat{p}_3)$. Because the leaky noisy-OR model [9] is utilized as the reasoning framework of the SBPN, $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$ can be derived based on $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$, $\Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2)$, and $\Pr(\hat{p}_6 | \bar{t}_1, \hat{t}_2)$, i.e., only three parameters are required to define $m(p_6)$. Clearly, the size of $m(p_6)$ is significantly smaller than that of a traditional CPT.

The method for obtaining the conditional probability of a place is formally presented as follows. For place p_j , let $S(\bullet p_j) = \bigcup_{t_i \in \bullet p_j} \{s(t_i)\}$ denote the states of all its input transitions, where $s(t_i)$ represents the state of transition t_i and satisfies $s(t_i) = \hat{t}_i$, $\bar{t}_i$, where $\hat{t}_i$ (or $\bar{t}_i$) indicates that t_i is (or is not) enabled. Let $\Pr(\hat{p}_j | S(\bullet p_j)) = \Pr(\hat{p}_j | \bigcup_{t_i \in \bullet p_j} \{s(t_i)\})$, where $s(t_i) = S(\bullet t_i)$.

Based on the above conditional probabilities of the transitions and places, the JPD of an SBPN can be derived. In fact, synergy effects can affect the conditional probability of a place, which is illustrated with propositions d_1 – d_6 in Fig. 2(a). Consider the following cases.

Case 1: d_1, d_2, d_4 , and d_5 hold, but d_3 does not. In this case, t_1 is enabled but t_2 is not. Moreover, t_1 is affected by the catalytic effect of p_4 , which increases the value of $\Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2)$. Mathematically, the strength of the catalytic effect can be quantified as $\lambda(p_4, t_1)$ ($0 < \lambda(p_4, t_1) \leq 1$). On this basis, $\Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2)$, which is affected by the catalytic effect, can be expressed as $\Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2)' = \Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2) + \Pr(\hat{p}_4) \lambda(p_4, t_1) (1 - \Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2))$.

Case 2: d_2 – d_5 hold, but d_1 does not. In this case, t_2 is enabled but t_1 is not. Moreover, t_2 is affected by the inhibitory effect of p_5 , which reduces the value of $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$. The strength of the inhibitory effect can be quantified as $\lambda(p_5, t_2)$ ($-1 \leq \lambda(p_5, t_2) < 0$). On this basis, $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$, which is affected by the inhibitory effect, can be expressed as $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)' = \Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2) - \Pr(\hat{p}_5) \lambda(p_5, t_2) \Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$.

Case 3: d_1 – d_6 hold. In this case, t_1 and t_2 are enabled. More importantly, t_1 is affected by the catalytic effect of p_4 and t_2 is affected by the inhibitory effect of p_5 . Accordingly, the value of $\Pr(\hat{p}_6 | \hat{t}_1, \hat{t}_2)$ is changed. It can be derived from the probabilities $\Pr(\hat{p}_6 | \hat{t}_1, \bar{t}_2)'$ and $\Pr(\bar{p}_6 | \hat{t}_1, \hat{t}_2)'$ if the reasoning idea of a leaky noisy-OR model is adopted, as discussed in Section 5.

In addition, a transition can be influenced by multiple synergy effects. The reasoning method for this case is presented in Section 5.

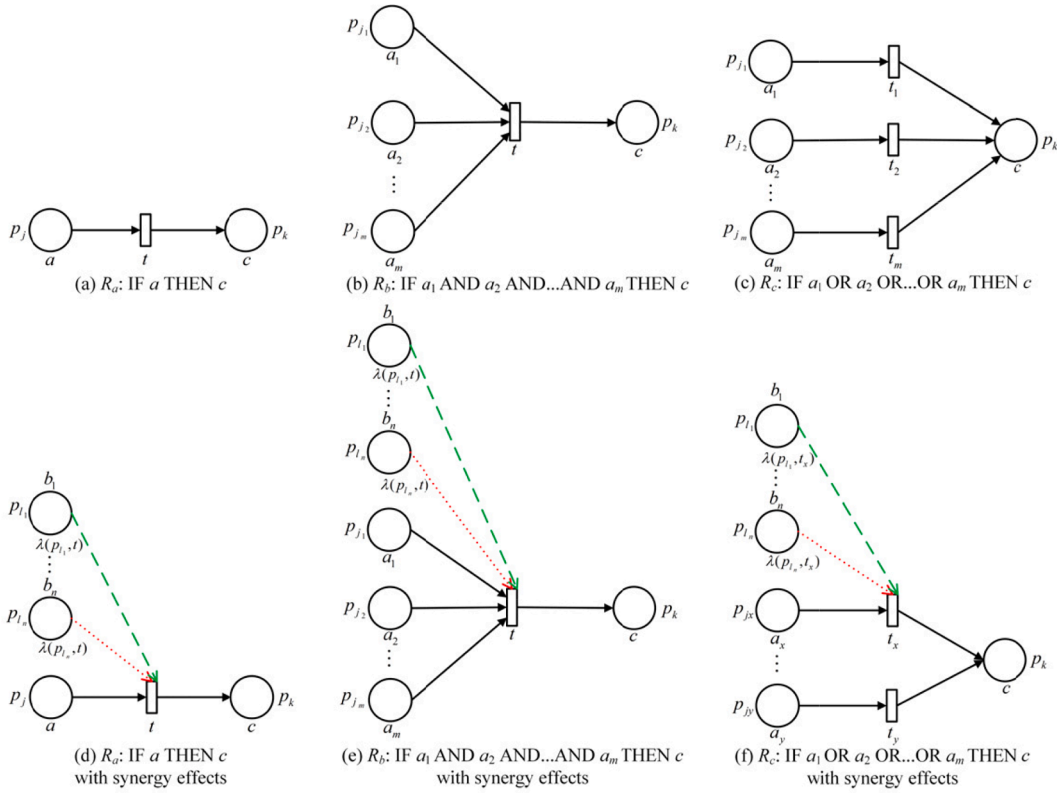


Fig. 3. Causal structures of the SBPN.

3.2. Causality representation with SBPN

This section introduces the causal structure of the SBPN, which is the basis of the knowledge mining and reasoning of the SBPN.

The basic causal structures of the SBPN are shown in Figs. 3(a)–(c). Specifically, Fig. 3(a) shows a simple rule that can be expressed as “ R_a : IF a THEN c ”. This rule indicates that if proposition a holds, proposition c holds. Fig. 3(b) gives a “conjunctive” rule with an “AND” structure among all causes, which can be expressed as “ R_b : IF a_1 AND a_2 AND ... AND a_m THEN c ”. This rule indicates that proposition c holds if propositions $a_1, a_2, \dots$, and a_m hold simultaneously. Fig. 3(c) shows a “disjunctive” rule with “OR” structures among all causes, which can be expressed as “ R_c : IF a_1 OR a_2 OR ... OR a_m THEN c ”. This rule indicates that if any of the propositions $a_1, a_2, \dots$ and a_m hold, proposition c holds. Actually, R_c is composed of m sub rules, i.e., “ R_{c1} : IF a_1 THEN c ”, “ R_{c2} : IF a_2 THEN c ”, ..., and “ R_{cm} : IF a_m THEN c ”.

The above three types of rules can be affected by synergy effects, as shown in Figs. 3(d)–(f). Concretely, Fig. 3(d) shows that although proposition $b_i (i = 1, 2, \dots, n)$ is not a necessary condition for the occurrence of c , the probability value of c deduced from the rule “ R_a : IF a THEN c ” increases (or decreases) under the catalytic (or inhibitory) effect of b_i on R_a . In this study, a green dashed (or red dotted) arc from place p_i to transition t_j is used to depict the catalytic (or inhibitory) effect of proposition $\beta(p_i)$ on the rule corresponding to t_j . Fig. 3(e) indicates that although $b_i (i = 1, 2, \dots, n)$ is not a necessary condition for the occurrence of c , the probability value of c deduced from the rule “ R_b : IF a_1 AND a_2 AND ... AND a_m THEN c ” increases (or decreases) under the catalytic (or inhibitory) effect of b_i on R_b . Fig. 3(f) shows that for the rule “ R_c : IF a_1 OR a_2 OR ... OR a_m THEN c ”, there exist a sub rule “ R_{cx} : IF a_x THEN c ” affected by synergy effects and a sub rule “ R_{cy} : IF a_y THEN c ” that is not affected by any synergy effect. For R_{cx} , the probability value of c deduced from it increases (or decreases) under the catalytic (or inhibitory) effect of $b_i (i = 1, 2, \dots, n)$ on it.

4. Knowledge mining of SBPN

4.1. SBPN structure mining

The structure mining of SBPN consists of the following three steps: 1) determine the antecedent-consequence relations among propositions; 2) identify the “AND/OR” relations among antecedent propositions; and 3) discover the synergy effects of propositions on rules. Consider for instance the chemical reactions a and b mentioned in Section 1. The process of mining the “AND/OR” relations and synergy effects existing in them is shown in Table 1. The mining algorithms corresponding to Steps 1–3 in Table 1 are discussed in detail in Sections 4.2–4.4, respectively.

Table 1
Instance of SBPN structure mining.

SBPN structure mining	Running example	Description
Step 1. Determine the antecedent-consequence relations among propositions		Determine the antecedent-consequence relations among propositions d_1 – d_6 , then we can find that there exist antecedent-consequence relations from d_1 , d_2 , and d_3 to d_6 .
Step 2. Identify the “AND/OR” relations among antecedent propositions		Identify the “AND/OR” relations among d_1 – d_3 , then we can get the following rules: “ R_1 : IF d_1 THEN d_6 ” and “ R_2 : IF d_2 AND d_3 THEN d_6 ”.
Step 3. Discover the synergy effects of propositions on rules		Discover the catalytic effect of d_4 on R_1 and the inhibitory effect of d_5 on R_2 .

4.2. Determining antecedent-consequence relation

The PC algorithm [24,25] and ANM [28,29] are two classical causal discovery methods. The PC algorithm aims to identify causal relations using a series of conditional independence tests and defined constraint rules with respect to causal directions. The ANM utilizes the asymmetry of noise in a causal relation to determine the causal direction between two variables. Because the PC algorithm can effectively mine the causality between discrete variables, this study uses it to discover common causal relations. Because it cannot always determine the causal direction of an edge, the ANM is employed to identify the causal direction of its generated undirected edges. This study adopts the Hilbert-Schmidt independence criterion [37] to test whether the variables and their corresponding residuals are statistically independent. The antecedent-consequence relation mining is realized via Algorithm 1.

Algorithm 1 Antecedent-consequence relation mining

Input: data samples ∂ of a variable set X , where each sample records all variables' values.

Output: a causal graph G_2 .

- 1: Learn a causal graph G_1 on ∂ using the PC algorithm;
- 2: **for** each undirected edge $x_i - x_j$ in G_1 **do**
- 3: Identify the causal order between the variables x_i and x_j via the ANM;
- 4: **end for**
- 5: Update G_1 to G_2 ;
- 6: **return** G_2

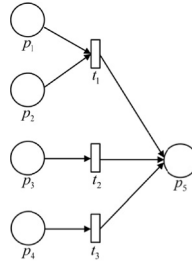


Fig. 4. “AND/OR” structure represented by a PN.

Table 2
Conditional probability w.r.t. p_5 .

		$\hat{p}_5$	$\tilde{p}_5$
$\hat{p}_1$	$\hat{p}_2$	0.71	0.29
$\tilde{p}_1$	$\tilde{p}_2$	0.74	0.26
$\hat{p}_1$	$\tilde{p}_2$	0.68	0.32
$\tilde{p}_1$	$\hat{p}_2$	0.95	0.05

4.3. Identifying “AND/OR” relation

Here, to save space, a rule “R: IF a AND b THEN c ” is abbreviated as “R: $a \wedge b \rightarrow c$ ”. Assume that proposition z is not affected by noise and has only two antecedent propositions x and y . Then, the principles for determining the “AND/OR” relation between x and y are as follows:

- 1) For rules $R_1: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, $R_2: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, $R_3: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, and $R_4: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, if R_1 – R_4 hold simultaneously, there is an “AND” relation between x and y , and vice versa.
- 2) For rules $R_1^*: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, $R_2^*: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, $R_3^*: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, and $R_4^*: \hat{x} \wedge \hat{y} \rightarrow \hat{z}$, if R_1^* – R_4^* hold simultaneously, there is an “OR” relation between x and y , and vice versa.

However, when z has antecedent propositions other than x and y , or is influenced by noise, determining the “AND/OR” relation between x and y according to the above principles has limitations. To illustrate this problem, this study randomly generates 1000 samples³ according to the causal model in Fig. 4. Concretely, let $\tau(v) \in \{0, 1\}$ denote the state value of variable v . For place p , $\tau(p)$ indicates whether proposition $\beta(p)$ is true; As for transition t , $\tau(t)$ denotes whether t is enabled. First, $\tau(p_1)$ – $\tau(p_4)$ are randomly taken as 0 or 1. Then, let $\tau(p_5) = \max\{\tau(t_1), \tau(t_2), \tau(t_3)\}$, where $\tau(t_1) = \min\{\tau(p_1), \tau(p_2)\}$, $\tau(t_2) = \tau(p_3)$ and $\tau(t_3) = \tau(p_4)$. To simulate noise, this study randomly selects 5 % of the samples where the values of p_5 are all set as $\tau(p_5) = 1 - \max\{\tau(t_1), \tau(t_2), \tau(t_3)\}$.

Table 2 provides the conditional probability of p_5 given p_1 and p_2 on the generated samples. As can be seen, there may be an “OR” relation between p_1 and p_2 , since Table 2 indicates that proposition $\beta(p_5)$ is true if either proposition $\beta(p_1)$ or $\beta(p_2)$ is true. However, as shown in Fig. 4, p_1 and p_2 actually have an “AND” relation. The reason that this “AND” relation cannot be identified from the data in Table 2 is as follows: p_5 is affected by the “OR” relations among t_1 , t_2 , and t_3 , which weakens the impact of the “AND” relation between p_1 and p_2 . To address this issue, based on conditional mutual information, an “AND strength” index is defined to effectively mine “AND” relations from data, as follows:

$$\begin{cases} \Phi(x, y, z) = 1/(1 + \exp(\omega_a - \omega_b)) \\ \omega_a = I(y; z|\hat{x}) \\ \omega_b = I(y; z|\tilde{x}) \end{cases} \quad (6)$$

Where x , y , and z are three propositions with antecedent-consequence relations from x and y to z . $I(y; z|\hat{x})$ (or $I(y; z|\tilde{x})$) denotes the quantity of information shared between y and z when x is false (or true). The difference between ω_a and ω_b can be used to measure the strength of an “AND” relation. To express it more intuitively, the difference is normalized by function $\Phi(x, y, z)$. Usually, the greater the values of $\Phi(x, y, z)$ and $\Phi(y, x, z)$, the greater the probability that there is an “AND” relation between x and y .

Theorem 1. For propositions x , y and z in a logical rule-based system, if 1) z is not affected by noise and has only two antecedent propositions x and y ; and 2) $\Pr(\hat{x}, \hat{y}) \neq 0$, $\Pr(\hat{x}, \tilde{y}) \neq 0$, $\Pr(\tilde{x}, \hat{y}) \neq 0$, and $\Pr(\tilde{x}, \tilde{y}) \neq 0$, then proposition d_1 : “ $\Phi(x, y, z) > 0.5$ ” (or

³ <https://github.com/netlearninglabs/SBPN/tree/main/sample>.

proposition d_2 : “ $\Phi(y, x, z) > 0.5$ ”) is a sufficient and necessary condition for proposition d_3 : “There is an “AND” relation between x and y ”.

Proof. First, we prove the sufficiency of this assertion. Suppose that d_1 and the negation of d_3 (i.e., $\neg d_3$: “There is an “OR” relation between x and y ”) can hold simultaneously. If the latter holds, we have $\Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = 0$, $\Pr(\hat{z}|\hat{x}) = 1$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, and $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$. On this basis, we obtain

$$\begin{aligned} I(y, z|\hat{x}) &= \sum_{y=\hat{y}, \hat{y}; z=\hat{z}, \hat{z}} \Pr(y, z|\hat{x}) \log \left[\frac{\Pr(y, z|\hat{x})}{\Pr(y|\hat{x})\Pr(z|\hat{x})} \right] = 0 \\ I(y, z|\hat{x}) &= \sum_{y=\hat{y}, \hat{y}; z=\hat{z}, \hat{z}} \Pr(y, z|\hat{x}) \log \left[\frac{\Pr(y, z|\hat{x})}{\Pr(y|\hat{x})\Pr(z|\hat{x})} \right] \\ &= \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] + \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] \\ &= \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] + (1 - \Pr(\hat{y}|\hat{x})) \log \left[\frac{1}{1 - \Pr(\hat{y}|\hat{x})} \right] \end{aligned}$$

Since $\Pr(\hat{y}, \hat{x}) \neq 0$ and $\Pr(\hat{y}, \hat{x}) \neq 0$, $\Pr(\hat{x}) = \Pr(\hat{y}, \hat{x}) + \Pr(\hat{y}, \hat{x}) > \Pr(\hat{y}, \hat{x}) > 0$. Therefore, $0 < \Pr(\hat{y}|\hat{x}) < 1$. Then, it can be derived that $I(y, z|\hat{x}) > 0$. Hence, $I(y, z|\hat{x}) > I(y, z|\hat{x})$. Since $f(x) = 1/(1 + \exp(x))$ is a monotonically decreasing function, when $I(y, z|\hat{x}) - I(y, z|\hat{x}) > 0$, $\Phi(x, y, z) = 1/(1 + \exp(I(y, z|\hat{x}) - I(y, z|\hat{x}))) < 1/(1 + \exp(0)) = 0.5$. However, this conclusion does not satisfy the above assumption. Hence, the assumption is not valid. In other words, if d_1 holds, then d_3 holds. The sufficiency is proven.

Next, we prove the necessity of the assertion. If d_3 holds, we have $\Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x}) = 0$, $\Pr(\hat{z}|\hat{x}) = 1$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$, and $\Pr(\hat{y}|\hat{x}) = \Pr(\hat{z}|\hat{x}) = \Pr(\hat{y}, \hat{z}|\hat{x})$. On this basis, we can get $I(y, z|\hat{x}) = 0$ and $I(y, z|\hat{x}) = \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] + \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] = \Pr(\hat{y}|\hat{x}) \log \left[\frac{1}{\Pr(\hat{y}|\hat{x})} \right] + (1 - \Pr(\hat{y}|\hat{x})) \log \left[\frac{1}{1 - \Pr(\hat{y}|\hat{x})} \right]$. Since $\Pr(\hat{y}, \hat{x}) \neq 0$ and $\Pr(\hat{y}, \hat{x}) \neq 0$, $\Pr(\hat{x}) = \Pr(\hat{y}, \hat{x}) + \Pr(\hat{y}, \hat{x}) > \Pr(\hat{y}, \hat{x}) > 0$. Therefore, $0 < \Pr(\hat{y}|\hat{x}) < 1$. Then, $I(y, z|\hat{x}) > 0$, i.e., $I(y, z|\hat{x}) > I(y, z|\hat{x})$. Consequently, $\Phi(x, y, z) > 0.5$, i.e., d_1 holds. The necessity is proven.

Similarly, it can be proven that d_2 is a sufficient and necessary condition for d_3 .

In fact, for a proposition with antecedent propositions, if it is not affected by noise, or there are both “AND” and “OR” structures among its antecedent propositions, Eq. (6) can still be used to determine the “AND/OR” relations among the antecedent propositions. Consider p_1 , p_2 , and p_5 in Fig. 4. We have $\Phi(p_1, p_2, p_5) > 0.5$ and $\Phi(p_2, p_1, p_5) > 0.5$, indicating that there is a high possibility of an “AND” relation between p_1 and p_2 .

On this basis, this study sets a threshold ϵ to extract “AND” relations from the data. Concretely, if $\Phi(x, y, z) > \epsilon$ and $\Phi(y, x, z) > \epsilon$, there may be an “AND” relation between x and y . Otherwise, there may be an “OR” relation between them. From Theorem 1, we obtain $0.5 \leq \epsilon < 1$. The larger ϵ , the stronger the discovered “AND” relation. In this paper, let $\epsilon = 0.5$.

The main idea of identifying “AND/OR” relations is as follows. Let $Parents(G_2, x_i)$ denote the cause set of variable x_i in causal graph G_2 learned by Algorithm 1. First, Eq. (6) is employed to measure the “AND strength” of each pair of variables x_j and x_k in $Parents(G_2, x_i)$. If it meets the above requirement of “AND” relation, a set $q_i = \{x_j, x_k\}$ is added to an “AND” relation set denoted by $AND(G_2, x_i)$. Each subset q_i of $AND(G_2, x_i)$, together with x_i , forms a rule “R: IF x_j AND x_k THEN x_i ”.

For a subset X_j of $Parents(G_2, x_i)$, if $\forall x_j, x_k \in X_j, \{x_j, x_k\} \subset AND(G_2, x_i)$, there is a composite “AND” relation among all elements in X_j . Subsequently, X_j is added to $AND(G_2, x_i)$, and all X_j ’s subsets contained in $AND(G_2, x_i)$ are removed from $AND(G_2, x_i)$. Consider the following case. Suppose that: 1) x_1 – x_7 are the variables in causal graph G_2 learned by Algorithm 1; and 2) $Parents(G_2, x_7) = \{x_1, x_2, \dots, x_6\}$ and $AND(G_2, x_7) = \{\{x_1, x_2\}, \{x_2, x_3\}, \{x_1, x_3\}, \{x_4, x_5\}\}$. As can be seen, there is a composite “AND” relation among x_1 , x_2 , and x_3 . Therefore, $AND(G_2, x_7)$ is updated to $AND(G_2, x_7) = \{\{x_1, x_2, x_3\}, \{x_4, x_5\}\}$, which can be expressed as the rules “R₁: IF x_1 AND x_2 AND x_3 THEN x_7 ” and “R₂: IF x_4 AND x_5 THEN x_7 ”. The maximum number of antecedent propositions of a rule is set as l_{max} . Consequently, for each subset Q_j of $AND(G_2, x_i)$, we have $2 \leq |Q_j| \leq l_{max}$, where $|Q_j|$ denotes the number of elements in Q_j . In this paper, let $l_{max} = 3$.

Finally, for each element x_j that belongs to $Parents(G_2, x_i)$ but not to $AND(G_2, x_i)$, it is integrated with $AND(G_2, x_i)$ to obtain a logical relation set denoted by $ANDOR(G_2, x_i)$. All elements in each subset of $ANDOR(G_2, x_i)$ form an “AND” structure, and the “AND” structures corresponding to all subsets of $ANDOR(G_2, x_i)$ interact with each other via “OR” relations. Consider variables x_1 – x_7 in the above-mentioned case, we have $Parents(G_2, x_7) = \{x_1, x_2, \dots, x_6\}$ and the updated $AND(G_2, x_7) = \{\{x_1, x_2, x_3\}, \{x_4, x_5\}\}$. After integrating x_6 with $AND(G_2, x_7)$, we can get $ANDOR(G_2, x_7) = \{\{x_1, x_2, x_3\}, \{x_4, x_5\}, \{x_6\}\}$, which corresponds to the rules “R₁: IF x_1 AND x_2

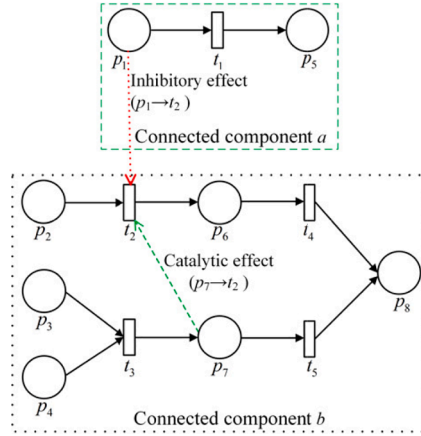


Fig. 5. Causal model represented by an SBPN.

AND x_3 THEN x_7 ", " R_2 : IF x_4 AND x_5 THEN x_7 ", and " R_3 : IF x_6 THEN x_7 ". The "AND/OR" relation mining is implemented via Algorithm 2.

Algorithm 2 "AND/OR" relation mining

Input: X : the set of variables in data samples ∂ , G_2 : causal graph of X learned by Algorithm 1, ϵ : threshold for determining "AND" relations, and l_{max} : the maximum number of causes in an "AND" relation.

Output: $\bigcup_{x_i \in X} \{ANDOR(G_2, x_i)\}$: logical relation sets of all variables X in G_2 .

```

1:  for  $\forall x_i \in X$  do
2:      Initialize  $AND(G_2, x_i) = \emptyset$  and  $ANDOR(G_2, x_i) = \emptyset$ ;
3:      /* Let  $|Parents(G_2, x_i)|$  denote the number of causes of variable  $x_i$  */
4:      if  $|Parents(G_2, x_i)| = 1$  then
5:           $ANDOR(G_2, x_i) = Parents(G_2, x_i)$ ;
6:      else
7:          if  $|Parents(G_2, x_i)| > 1$  then
8:              for  $\forall x_j, x_k \in Parents(G_2, x_i)$  do
9:                  if  $\Phi(x_j, x_k, x_i) \geq \epsilon$  and  $\Phi(x_k, x_j, x_i) \geq \epsilon$  then
10:                     Add  $q_i = \{x_j, x_k\}$  to  $AND(G_2, x_i)$ ;
11:                 end if
12:             end for
13:             if  $l_{max} \geq 3$  then
14:                 for  $l = 3$  to  $l_{max}$  do
15:                     for  $\forall X_j \subseteq Parents(G_2, x_i) \wedge |X_j| = l$  do
16:                         if  $\forall x_j, x_k \in X_j, \{x_j, x_k\} \subset AND(G_2, x_i)$  then
17:                             Add  $X_j$  to  $AND(G_2, x_i)$ ;
18:                         end if
19:                     end for
20:                 end for
21:                 for  $\forall Q_i, Q_j \subset AND(G_2, x_i) \wedge Q_i \subset Q_j$  do
22:                     Remove  $Q_i$  from  $AND(G_2, x_i)$ ;
23:                 end for
24:             end if
25:             for  $\forall x_j \in Parents(G_2, x_i) \cap AND(G_2, x_i)$  do
26:                 Remove  $x_j$  from  $Parents(G_2, x_i)$ ;
27:             end for
28:              $ANDOR(G_2, x_i) = Parents(G_2, x_i) \cup AND(G_2, x_i)$ ;
29:         end if
30:     end if
31: end for

```

4.4. Discovering synergy effect

The synergy effect that this study aims to mine is associated with the PN transition. In other words, the PN transition is the basic unit of synergy effect mining. In fact, each variable x_i in causal graph G_2 learned by Algorithm 1 can be associated with a place p_i such that $\beta(p_i) = x_i$. Accordingly, each logical relation set $ANDOR(G_2, x_i)$ generated by Algorithm 2 can be mapped to a set $ANDOR(G_2, p_i)$ such that $\forall x_j \in ANDOR(G_2, x_i) \exists p_j \in ANDOR(G_2, p_i) : \beta(p_j) = x_j$. Each subset $Q_i \subset ANDOR(G_2, p_i)$ can be associated with a transition t_i satisfying that $\bullet t_i = Q_i$ and $t_i^* = \{p_i\}$. Consider, for instance, the causal model in Fig. 5, where 1) p_1 – p_8 are places associated with a causal graph G_2 learned by Algorithm 1; and 2) $ANDOR(G_2, p_5) = \{\{p_1\}\}$, $ANDOR(G_2, p_6) = \{\{p_2\}\}$, $ANDOR(G_2, p_7) = \{\{p_3, p_4\}\}$, and

$ANDOR(G_2, p_8) = \{\{p_6\}, \{p_7\}\}$. Then, transitions t_1 – t_5 can be defined, which satisfy: $\bullet t_1 = \{p_1\}, t_1^* = \{p_5\}, \bullet t_2 = \{p_2\}, t_2^* = \{p_6\}, \bullet t_3 = \{p_3, p_4\}, t_3^* = \{p_7\}, \bullet t_4 = \{p_6\}, t_4^* = \{p_8\}, \bullet t_5 = \{p_7\}$, and $t_5^* = \{p_8\}$.

Given an SBPN, let $CausalityPlace(t_i) = \{p_j | (p_j, t_i) \in F^* \vee (t_i, p_j) \in F^*\}$ be the set of places that can be connected with t_i through solid arcs, where $p_j \in P, t_i \in T$, and “ $*$ ” means closure operation. Considering t_2 in Fig. 5, we obtain $CausalityPlace(t_2) = \{p_2, p_6, p_8\}$. Neglecting the directions of all solid arcs indicating causality, and removing all dashed/dotted arcs indicating synergy effects in an SBPN, we can obtain an undirected graph, where the set of places in the same connected component as t_i is denoted as $ConnectPlace(t_i)$. Considering t_2 in the connected component b of the SBPN in Fig. 5, we obtain $ConnectPlace(t_2) = \{p_2, p_3, p_4, p_6, p_7, p_8\}$.

Let $UP(t_i) = P - ConnectPlace(t_i)$ be a set of potential synergy places. It is used to discover the synergy effect imposed on t_i by a place that is not in the same connected component as t_i , which is called an inter-structure synergy (e.g., the inhibitory effect of p_1 on t_2 shown in Fig. 5). Let $RP(t_i) = ConnectPlace(t_i) - CausalityPlace(t_i)$ also be a set of potential synergy places. Different from $UP(t_i)$, $RP(t_i)$ is utilized to mine the synergy effect imposed on t_i by the place in the same connected component as t_i (such that it has no direct/indirect causality with t_i), which is called the intra-structure synergy (e.g., the catalytic effect of p_7 on t_2 shown in Fig. 5). The set $RP(t_i) \cup UP(t_i)$ can then be used to capture each synergy effect imposed on t_i .

Let $PotentSynergy(t_i)$ be a set of places that may have synergy effects on t_i , which can be derived from $RP(t_i)$ or $UP(t_i)$. Concretely, when a user’s need is “inter-structure synergy mining”, $PotentSynergy(t_i) = UP(t_i)$; when it is “intra-structure synergy mining”, $PotentSynergy(t_i) = RP(t_i)$; when it is “synergy mining”, $PotentSynergy(t_i) = RP(t_i) \cup UP(t_i)$.

To mine the synergy effects associated to t_i , for each place p_j in $PotentSynergy(t_i)$, the strength of its synergy effect imposed on t_i should be measured. A natural way is to directly calculate the difference between $\Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}, \hat{p}_j\right)$ and $\Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}\right)$. Unfortunately, incorrect antecedent-consequence relations learned from data can affect the effectiveness of identifying synergy effects. To address this issue, a filter function, $y = \tan(k(x - 0.5)) (0 < k < \pi)$, is presented to capture the synergy effects imposed on a rule with high credibility. Based on this, this study proposes a method for measuring the strength of the synergy effect as

$$\begin{cases} \Theta(t_i, p_j) = 1/(1 + \exp(\sigma_a - \sigma_b)) \\ \Psi(t_i, p_j) = 1/(1 + \exp(\sigma_b - \sigma_a)) \\ \sigma_a = \tan\left(k\left(\Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}\right) - 0.5\right)\right) \\ \sigma_b = \tan\left(k\left(\Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}, \hat{p}_j\right) - 0.5\right)\right) \\ \text{s.t.} \\ 0 < k < \pi \end{cases} \quad (7)$$

where functions $\Theta(t_i, p_j)$ and $\Psi(t_i, p_j)$ are used to measure the strengths of the catalytic and inhibitory effects, respectively. The larger the value of $\Theta(t_i, p_j)$ (or $\Psi(t_i, p_j)$), the stronger the catalytic (or inhibitory) effect of p_j on t_i .

Theorem 2. If p_j has a catalytic (or inhibitory) effect on t_i , then $\Theta(t_i, p_j) > 0.5$ (or $\Psi(t_i, p_j) > 0.5$).

Proof. Let $e_1 = \Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}\right)$ and $e_2 = \Pr\left(\bigcup_{p_m \in t_i^*} \{\hat{p}_m\} \middle| \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}, \hat{p}_j\right)$. If p_j has a catalytic effect on t_i , $e_2 > e_1$. Since $0 \leq e_1, e_2 \leq 1$ and $0 < k < \pi$, we have $-\pi/2 < k(e_1 - 0.5) < k(e_2 - 0.5) < \pi/2$. Because the function $g(x) = \tan(x)$ monotonically increases over $x \in (-2/\pi, 2/\pi)$, $\sigma_b = \tan(k(e_2 - 0.5)) < \sigma_a = \tan(k(e_1 - 0.5))$. On this basis, we can obtain $\Theta(t_i, p_j) = 1/(1 + \exp(\sigma_b - \sigma_a)) > 1/(1 + \exp(0)) = 0.5$. Similarly, it can be proven that if p_j has an inhibitory effect on t_i , $\Psi(t_i, p_j) > 0.5$.

The thresholds for identifying catalytic and inhibitory effects are set as γ and δ , respectively. From Theorem 2, we obtain $0.5 < \gamma < 1$ and $0.5 < \delta < 1$. Let $CatalytEffect(t_i)$ (or $InhibitEffect(t_i)$) be a set of places, where each place $p_j (p_j \in PotentSynergy(t_i))$ has a catalytic (or inhibitory) effect on t_i . Concretely, if $\Theta(t_i, p_j) \geq \gamma$, p_j has a catalytic effect on t_i and is added to $CatalytEffect(t_i)$. If $\Psi(t_i, p_j) \geq \delta$, p_j has an inhibitory effect on t_i and is added to $InhibitEffect(t_i)$. Consider t_2 in Fig. 5, in a “synergy mining” pattern, $p_7 \in CatalytEffect(t_2)$ and $p_1 \in InhibitEffect(t_2)$. In this study, the parameter k in (7) is set to $2\pi/3$.

Using the above concepts, the input, output, and synergy places of all defined transitions can be obtained, resulting in an SBPN structure. The specific synergy effect mining is realized via Algorithm 3.

Algorithm 3 Synergy effect mining

Input: P : the set of places corresponding to all variables X in data samples ∂ ; G_2 : causal graph of X learned by Algorithm 1, $\bigcup_{p_i \in P} \{ANDOR(G_2, p_i)\}$: logical relation sets of all places P , γ and δ : thresholds for identifying synergy effects, and “user’s need”.

Output: G_3 : an SBPN structure.

```

1: for  $\forall p_i \in P$  do
2:   for  $\forall Q_i \subseteq ANDOR(G_2, p_i)$  do
3:     Let  $t_i$  be a transition such that  $\bullet t_i = Q_i$  and  $t_i^* = \{p_i\}$ ;
4:     Add  $t_i$  to  $G_2$  and create directed arcs from  $\bullet t_i$  to  $t_i$  and  $t_i$  to  $t_i^*$ ;
5:     if “user’s need” == “inter-structure synergy mining” then
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(continued on next page)

(continued)

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6:      PotentSynergy( $t_i$ ) =  $UP(t_i)$ ;
7:      end if
8:      if “user’s need” == “intra-structure synergy mining” then
9:          PotentSynergy( $t_i$ ) =  $RP(t_i)$ ;
10:      end if
11:      if “user’s need” == “synergy mining” then
12:          PotentSynergy( $t_i$ ) =  $RP(t_i) \cup UP(t_i)$ ;
13:      end if
14:      Initialize  $CatalytEffect(t_i) = \emptyset$  and  $InhibitEffect(t_i) = \emptyset$ ;
15:      for  $\forall p_j \in PotentSynergy(t_i)$  do
16:          if  $\Theta(t_i, p_j) \geq \gamma$  then
17:              Add  $p_j$  to  $CatalytEffect(t_i)$ ;
18:          else
19:              if  $\Psi(t_i, p_j) \geq \delta$  then
20:                  Add  $p_j$  to  $InhibitEffect(t_i)$ ;
21:              end if
22:          end if
23:      end for
24:      Create dashed (or dotted) arc(s) from  $CatalytEffect(t_i)$  (or  $InhibitEffect(t_i)$ ) to  $t_i$ ;
25:  end for
26: end for
27: Update  $G_2$  to  $G_3$ ;
28: return  $G_3$ 

```

5. Probabilistic reasoning of SBPN

This section introduces the probabilistic reasoning underlying the SBPN. As discussed in Section 3.1, synergy effects can affect the conditional probability of a place given its input transitions. This issue is illustrated next.

The leaky noisy-OR model [15], which is a BN-based probabilistic reasoning method, can simplify the CPT using causal independence hypotheses. Based on this, this study presents a probabilistic reasoning framework with transitions as the basic reasoning elements:

$$\Pr(\hat{p}_x | S(\bullet p_x)) = 1 - (1 - \Pr(\hat{p}_x | \tilde{t}_1, \dots, \tilde{t}_n)) \prod_{j \in J} (1 - \Pr(\hat{p}_x | \tilde{t}_1, \dots, \tilde{t}_{j-1}, \hat{t}_j, \tilde{t}_{j+1}, \dots, \tilde{t}_n)) \quad (8)$$

where $\hat{p}_x$ indicates that the proposition $\beta(p_x)$ holds, $S(\bullet p_x)$ denotes the states of all input transitions of p_x , J denotes the indices of the enabling transitions in $S(\bullet p_x)$, and $\hat{t}_j$ indicates that the transition t_j ($t_j \in \bullet p_x$) is enabled. t_j can be simultaneously affected by both catalytic and inhibitory effects. In this case, the value of $\Pr(\hat{p}_x | \tilde{t}_1, \dots, \tilde{t}_{j-1}, \hat{t}_j, \tilde{t}_{j+1}, \dots, \tilde{t}_n)$, denoted by r_j , changes as follows:

$$r_j = \left(\sum_{i=1}^m (r_j + \Pr(\hat{p}_i^+) \lambda(p_i^+, t_j) (1 - r_j)) + \sum_{j=1}^n (r_j - \Pr(\hat{p}_j^-) |\lambda(p_j^-, t_j)| r_j) \right) / (m + n)$$

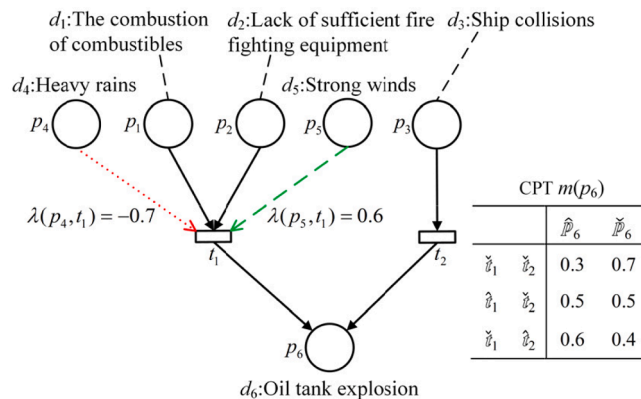


Fig. 6. Oil tank explosion case modelled by an SBPN.

$$\text{s.t.} \begin{cases} 0 < \lambda(p_i^+, t_j) \leq 1 \\ -1 \leq \lambda(p_j^-, t_j) < 0 \\ p_i^+ \neq p_j^- \end{cases} \quad (9)$$

In (9), p_i^+ and p_j^- represent a catalytic place and an inhibitory one of t_j , respectively. $\lambda(p_i^+, t_j)$ and $\lambda(p_j^-, t_j)$ are the synergy coefficients of p_i^+ and p_j^- with respect to t_j , respectively, which can be learned from data or given based on expert experience. Additionally, m and n are the numbers of catalytic and inhibitory places of t_j , respectively.

Next, a risk assessment case of an oil tank explosion, shown in Fig. 6, is used to illustrate how to carry out probabilistic reasoning considering synergy effects. Assume that the CPT $m(p_6)$ and synergy coefficients $\lambda(p_4, t_1)$ and $\lambda(p_5, t_1)$ in Fig. 6 are provided by the experts. Using (8) and (9), the reasoning results for place p_6 in the following cases can be derived.

Case 1: d_1, d_2 and d_3 are true, whereas d_4 and d_5 are false. In this case, t_1 and t_2 are enabled. By using (8) and CPT $m(p_6)$ in Fig. 6, we have $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) = 1 - (1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2))(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2))(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.86$.

Case 2: d_1, d_2, d_3 , and d_4 are true, whereas d_5 is false. Clearly, t_1 and t_2 are enabled. More importantly, p_4 has an inhibitory effect on t_1 , which reduces the value of $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)$. Concretely, by using (9) and $m(p_6)$, we have $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)' = \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) - \Pr(\hat{p}_4|\lambda(p_4, t_1)|\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.5 - 1 \times 0.7 \times 0.5 = 0.15$. According to (8), $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) = 1 - (1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2))(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)')(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.76$.

Case 3: d_1, d_2, d_3 , and d_5 are true, whereas d_4 is false. Clearly, t_1 and t_2 are enabled. Moreover, p_5 has a catalytic effect on t_1 , which increases the value of $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)$. By using (9), we have $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)'' = \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) + \Pr(\hat{p}_5)\lambda(p_5, t_1)(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.5 + 1 \times 0.6 \times (1 - 0.5) = 0.8$. According to (8), $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) = 1 - (1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2))(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)')(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.94$.

Case 4: d_1 – d_5 are all true. In this case, t_1 and t_2 are enabled. Because of the catalytic effect of p_5 on t_1 and the inhibitory effect of p_4 on t_1 , the value of $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)$ can change. Specifically, using (9), we obtain $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)''' = (\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)' + \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)'')/2 = 0.48$. According to (8), $\Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2) = 1 - (1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2))(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)''')(1 - \Pr(\hat{p}_6|\hat{t}_1, \hat{t}_2)) = 0.85$.

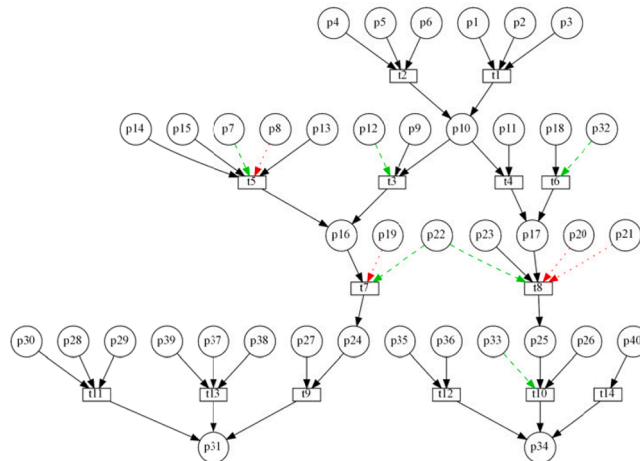
Quantifying synergy coefficients based on expert experience is challenging, particularly when the causal structure is complex. To address this issue, this study presents a method for learning synergy coefficients from the data. For synergy proposition x of rule R , the strength of the synergy effect of x on R is first learned from the data. The credibility of R considering the occurrence of x is utilized as the benchmark synergy strength of x on R . Then, this strength is fine-tuned based on the dataset, such that R has the highest credibility when all of its synergy propositions are considered. Finally, based on the mapping between the synergy strength and synergy coefficient discussed in Section 3.1, the synergy coefficient of x on R can be derived. More details on how to learn the synergy coefficients from the data can be found in the [supplementary material](#).

6. Experimental evaluation

This section presents three experiments and their results. The first and second experiments are conducted to evaluate the effectiveness of the SBPN in mining “AND/OR” relations and synergy effects on synthetic and real-world structures, respectively. The third experiment aims to assess the probabilistic reasoning performance of the SBPN in the presence of synergy effects.

6.1. Performance in mining “AND/OR” relations and synergy effects on synthetic structures

This study focuses on the “AND/OR” relation between causes and the synergy effect of a non-causal variable on a reasoning rule.



They depict a complex local causal structure that is composed of an effect variable, its direct causal variables and synergy factors. To evaluate the performance of the proposed model in mining “AND/OR” relations and synergy effects, various local causal structures with different numbers of “AND” structures, “OR” structures, and synergy effects should be constructed. To carry out evaluation efficiently and present the evaluation results compactly, the local causal structures are randomly integrated into some global ones, which are employed as the benchmark networks for mining “AND/OR” relations and synergy effects. Among them, the benchmark network shown in Fig. 7 has the largest number of “AND” structures and synergy effects. Considering its representativeness, this study utilizes it to discuss the performance of the proposed model in knowledge mining. The [supplementary material](#) presents the performance of the proposed model with respect to other representative benchmark networks.

The data generation mechanism is as follows. For a place without an input transition, its value is randomly set to zero or one. For a place p_m with input transitions $t_a, t_b, \dots$, and t_z , let

$$p_m = \left| u \left(1 - \left(1 - \frac{\sum_{i=1}^q p_{a_i} + q \prod_{i=1}^q p_{a_i}}{2q} \right) \times \left(1 - \frac{\sum_{i=1}^r p_{b_i} + r \prod_{i=1}^r p_{b_i}}{2r} \right) \times \dots \times \left(1 - \frac{\sum_{i=1}^s p_{z_i} + s \prod_{i=1}^s p_{z_i}}{2s} \right) \right) + \varepsilon_m \right|$$

$$\text{s.t. } u(x) = \begin{cases} 1 & x > 0.5 \\ 0 & x \leq 0.5 \end{cases} \quad (10)$$

where p_m, p_{a_i}, p_{b_i} , and p_{z_i} are the values of $p_m, p_{a_i} \in \bullet t_a, p_{b_i} \in \bullet t_b$, and $p_{z_i} \in \bullet t_z$, respectively, and $\varepsilon_m = -1, 0$ is a noise variable. Each fraction in (10) represents the impact of the input transition of p_m on p_m . The value of p_m is determined by the absolute value of all the impacts of its input transitions on it plus noise. The generation mechanism of the “AND” relations between causal variables is depicted by the product of the values of all the causal variables pointing to the same transition as well as the proportion of the causal variables that are true among all the causal variables. The noisy-OR model [13] is employed as the generation mechanism of the “OR” relations between transitions. Then, for a transition t_i affected by synergy effects, the values of its catalytic place p_i and inhibitory place p_j are updated such that $\Pr(\bigcup_{p_m \in \bullet t_i} \{\hat{p}_m\} | \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}, \hat{p}_i) \gg \Pr(\bigcup_{p_m \in \bullet t_i} \{\hat{p}_m\} | \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\})$ and $\Pr(\bigcup_{p_m \in \bullet t_i} \{\hat{p}_m\} | \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\}, \hat{p}_j) \ll \Pr(\bigcup_{p_m \in \bullet t_i} \{\hat{p}_m\} | \bigcup_{p_n \in \bullet t_i} \{\hat{p}_n\})$.

Based on the benchmark network shown in Fig. 7 and the data generation mechanism, this study randomly generated six datasets⁴ with different sample sizes and noise proportions. In this paper, a synthetic dataset is named in the form of “ $D_{x,y}$ ”, where “ D ” means a dataset, “ x ” represents a noise proportion ($x = 5, 10$), and “ y ” indicates a sample size ($y = 1000, 2000, 3000$). Consider a dataset named D_{5-1000} , it has 1000 data samples with noise accounting for 5 % of the samples.

The performance of the proposed SBPN is analyzed for different datasets. Consider for instance D_{5-1000} and $D_{10-3000}$. Their corresponding mining results are given in Figs. 8 and 9, respectively, where “ $\times$ ” indicates an incorrect synergy effect or antecedent-

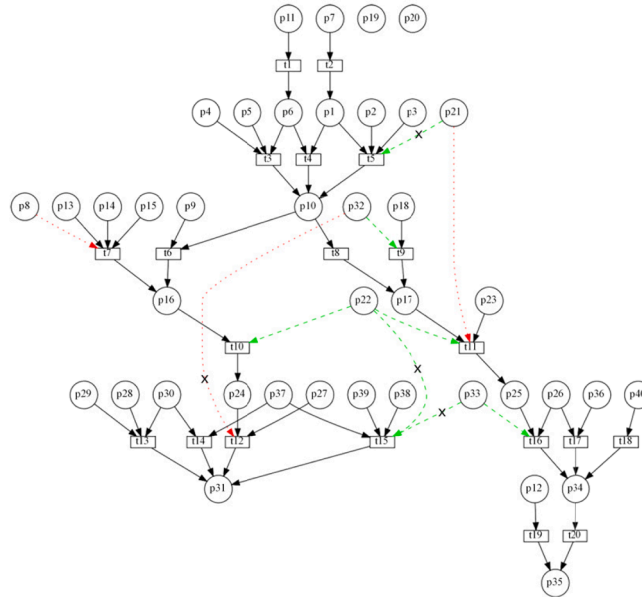


Fig. 8. Mining result of SBPN on D_{5-1000} dataset.

⁴ https://github.com/netlearninglabs/SBPN/tree/main/simulated_structure.

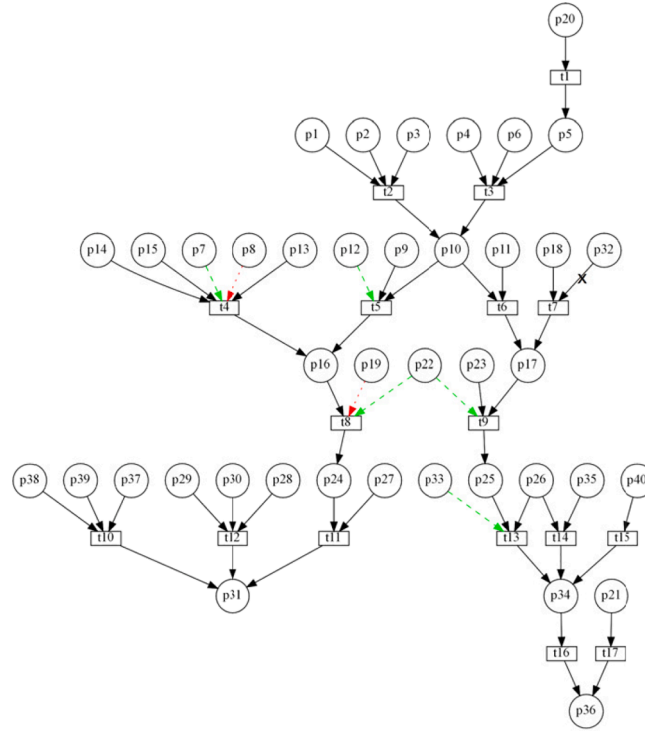


Fig. 9. Mining result of SBPN on $D_{10-3000}$ dataset.

consequence relation. Most of the “AND/OR” structures and synergy effects in Fig. 7 are correctly identified on the two datasets. Specifically, for D_{5-1000} , four incorrect synergy effects are identified, accounting for 40 % of the total number of synergy effects identified. In addition, no incorrect synergy effects are identified on $D_{10-3000}$. In terms of synergy effect mining, the performance of the proposed SBPN on D_{5-1000} is the worst. The reasons for this are as follows: A low percentage of noise and small number of samples can reduce the amount of synergy information in a synergy place. Additionally, it can be seen from Fig. 7 that p_{32} is actually a synergy place of rule “R: IF p_{18} THEN p_{17} ”. However, as shown in Fig. 9, an incorrect antecedent-consequence relation from p_{32} to p_{17} is identified on $D_{10-3000}$. The reasons for this are as follows: p_{32} carries a large amount of synergy information regarding R, resulting in a strong correlation between it and p_{17} .

To quantitatively analyze the performance of SBPN in mining “AND” relations, the concept of a minimal “AND” unit is defined. For a transition t_j with an output place p_x , if t_j has multiple input places $\bullet t_j (|\bullet t_j| > 1)$, each pair of places p_i and p_j that belong to $\bullet t_j$ can form a minimal “AND” unit toward p_x , denoted as $p_i \& p_j \rightarrow p_x$. Consider p_{13} – p_{16} in Fig. 7. They have the following three minimal “AND” units: $p_{13} \& p_{14} \rightarrow p_{16}$, $p_{13} \& p_{15} \rightarrow p_{16}$, and $p_{14} \& p_{15} \rightarrow p_{16}$. Moreover, each place p_i that belongs to $\bullet t_j$ can form a common causal relation toward p_x , denoted as $p_i \rightarrow p_x$. As seen in Fig. 7, the benchmark network has 21 minimal “AND” units, 10 synergy effects, and 30 common causal relations. Then, the following nine metrics are utilized to quantitatively evaluate the performance of SBPN in mining “AND/OR” relations and synergy effects:

- 1) Recall(C): Proportion of correctly detected common causal relations to the total common causal relations in the benchmark network;
- 2) Precision(C): Proportion of correctly detected common causal relations to the total common causal relations in the generated network;
- 3) F1(C): The harmonic average of Recall(C) and Precision(C);
- 4) Recall(A): Proportion of correctly detected minimal “AND” units to the total minimal “AND” units in the benchmark network;
- 5) Precision(A): Proportion of correctly detected minimal “AND” units to the total minimal “AND” units in the generated network;
- 6) F1(A): The harmonic average of Recall(A) and Precision(A);
- 7) Recall(S): Proportion of correctly detected synergy effects to the total synergy effects in the benchmark network;
- 8) Precision(S): Proportion of correctly detected synergy effects to the total synergy effects in the generated network; and
- 9) F1(S): The harmonic average of Recall(S) and Precision(S).

The performance of SBPN in mining “AND/OR” relations and synergy effects on all synthetic datasets can now be evaluated, as shown in Table 3. From columns “Recall(A)”, “Precision(A)”, and “F1(A)”, the proposed SBPN correctly identifies most of the “AND” relations in Fig. 7 and mistakenly captures a small number of “AND” relations on all datasets. From columns “Recall(S)”, “Precision

Table 3

Performance of SBPN in mining “AND/OR” relations and synergy effects.

Dataset	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
D_{5-1000}	0.93	0.88	0.90	0.90	0.86	0.88	0.60	0.60	0.60
D_{5-2000}	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.89	0.84
D_{5-3000}	1.00	0.97	0.98	1.00	0.95	0.97	0.60	1.00	0.75
$D_{10-1000}$	0.93	0.90	0.91	0.90	0.86	0.88	0.90	0.90	0.90
$D_{10-2000}$	0.97	0.85	0.91	0.90	0.90	0.90	0.70	0.88	0.78
$D_{10-3000}$	0.97	0.88	0.92	0.95	0.91	0.93	0.70	1.00	0.82

Table 4Performance in mining “AND/OR” relations and synergy effects on D_{5-1000} dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	0.93	0.88	0.90	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.16	0.16	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.16	0.16	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.67	0.57	0.62	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.70	0.82	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.60	0.36	0.45	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	0.93	0.88	0.90	0.90	0.86	0.88	0.60	0.60	0.60

Table 5Performance in mining “AND/OR” relations and synergy effects on D_{5-2000} dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	1.00	1.00	1.00	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.17	0.17	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.17	0.17	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.87	0.76	0.81	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	0.93	0.80	0.86	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.57	0.31	0.40	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.89	0.84

Table 6Performance in mining “AND/OR” relations and synergy effects on D_{5-3000} dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	1.00	0.97	0.98	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.16	0.16	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.16	0.16	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.70	0.66	0.68	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.88	0.94	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.57	0.32	0.41	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	1.00	0.97	0.98	1.00	0.95	0.97	0.60	1.00	0.75

(S)”, and “F1(S)”, it identifies most of the synergy effects in Fig. 7 and captures a small number of incorrect synergy effects on all datasets except D_{5-1000} . From columns “Recall(C)”, “Precision(C)”, and “F1(C)”, it can effectively discover common causal relations from the data. To verify the advantage of the SBPN in knowledge mining, a series of comparison experiments are conducted on the six datasets. The SBPN is compared with state-of-the-art causal discovery methods, including PC [24], ANM [28], CCI [17], HC [47], GES [45], and GOBNILP [46]. PC, HC, GES, and GOBNILP aim to discover global causal structures, whereas ANM and CCI focus on identifying local causal structures. In particular, the CCI can capture the combined effects among non-causal variables and among causal and non-causal variables. To ensure fairness in the comparison, all competing algorithms are given the same input data as the SBPN. For ANM and CCI, a series of conditional independence tests are first performed on all observed variables to determine dependencies, and then they are employed to discover causal relations from the dependencies.

The performances of all mining methods on the synthetic datasets are listed in Tables 4–9. From Tables 4–9, the PC and SBPN perform best in mining common causal relations because they can effectively capture the causal relations between discrete variables. In

Table 7Performance in mining “AND/OR” relations and synergy effects on $D_{10-1000}$ dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	0.93	0.90	0.91	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.17	0.17	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.17	0.17	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.23	0.20	0.21	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.86	0.92	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.57	0.35	0.43	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	0.93	0.90	0.91	0.90	0.86	0.88	0.90	0.90	0.90

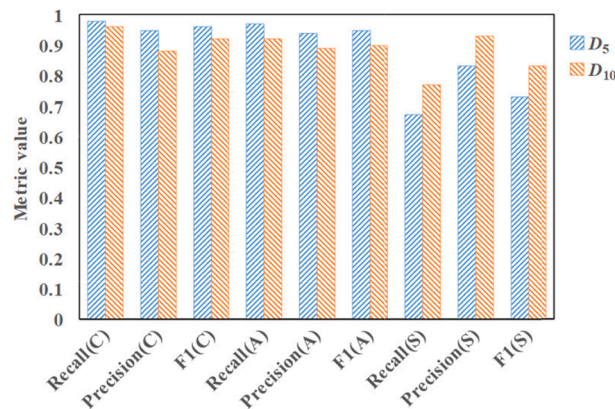
Table 8Performance in mining “AND/OR” relations and synergy effects on $D_{10-2000}$ dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	0.97	0.85	0.91	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.15	0.16	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.15	0.16	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.80	0.71	0.75	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.73	0.84	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.43	0.30	0.35	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	0.97	0.85	0.91	0.90	0.90	0.90	0.70	0.88	0.78

Table 9Performance in mining “AND/OR” relations and synergy effects on $D_{10-3000}$ dataset.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	0.97	0.88	0.92	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.19	0.18	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.19	0.18	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.40	0.36	0.38	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.81	0.90	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.60	0.33	0.43	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	0.97	0.88	0.92	0.95	0.91	0.93	0.70	1.00	0.82

addition, SBPN performs better than its six peers in capturing “AND/OR” relations and synergy effects. The reasons for this are as follows: PC, ANM, HC, GES, and GOBNILP focus on discovering common causal relations but fail to consider the complex causality containing “AND” relations as well as synergy effects. CCI can mine common causal relations, the “AND” relations among non-causal variables, and the enhancement effects among causal and non-causal variables. However, it is unable to capture the “AND” relations among causes and the catalytic or inhibitory effect that a non-causal variable may have on a reasoning rule (e.g., the catalytic effect of p_{33} on t_{10} shown in Fig. 7). In contrast, SBPN can not only discover the common causal relations, but also capture the “AND” relations

**Fig. 10.** Performance of SBPN in knowledge mining under different noise proportions.

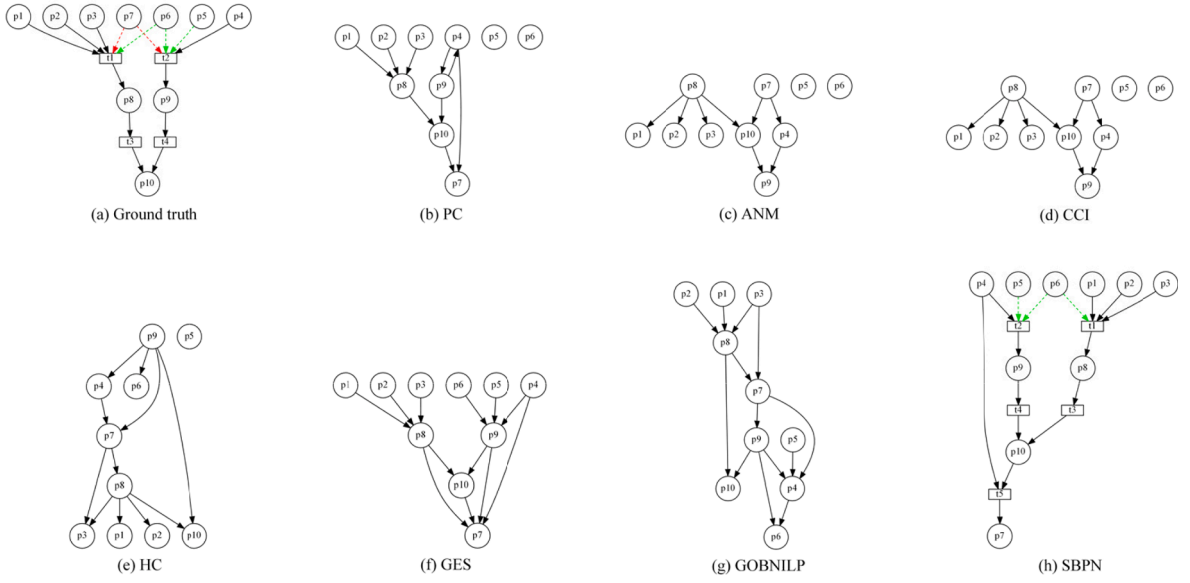


Fig. 11. Causal structures discovered from Case 1.

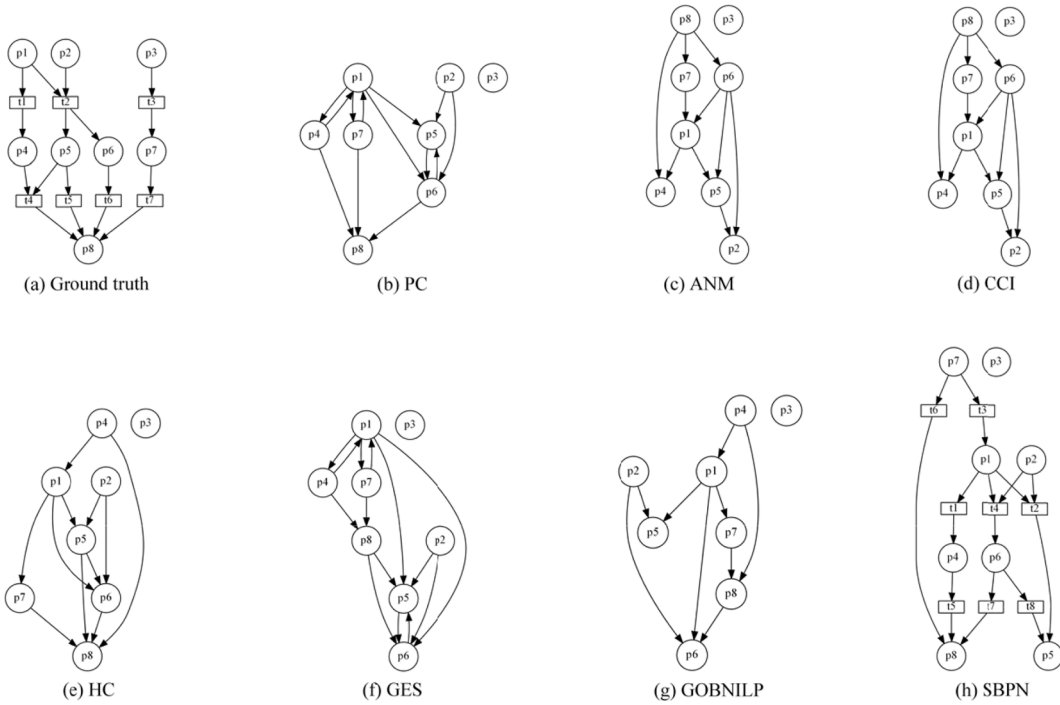


Fig. 12. Causal structures discovered from Case 2.

among causes and the catalytic or inhibitory effects.

To study the influence of the noise proportions on knowledge mining, the evaluation results in Table 3 are integrated based on different noise proportions. Specifically, for each column in Table 3, the results shown in Fig. 10 are obtained by taking the average of the evaluation results on the datasets with the same noise proportions. As can be seen from the columns “Recall(A)”, “Precision(A)”, and “F1(A)” in Fig. 10, in terms of identifying “AND” relations, the performance of the proposed SBPN on the datasets with 5 % noise (denoted by D_5) is better than that on the datasets with 10 % noise (denoted by D_{10}). This is because a low noise proportion allows a consequent proposition to contain more logical information regarding its antecedent propositions. Thus, an “AND” relation among antecedent propositions is easier to identify. As can be seen from the columns “Recall(S)”, “Precision(S)”, and “F1(S)” in Fig. 10, the

Table 10

Performance in mining “AND/OR” relations and synergy effects on Case 1.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	1.00	0.67	0.80	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.17	0.13	0.15	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.17	0.13	0.15	N.A.	N.A.	N.A.	0.00	N.A.	N.A.
HC	0.33	0.18	0.23	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	1.00	0.50	0.67	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.83	0.38	0.52	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	1.00	0.75	0.86	1.00	0.75	0.86	0.60	1.00	0.75

Table 11

Performance in mining “AND/OR” relations and synergy effects on Case 2.

Model	Metrics								
	Recall(C)	Precision(C)	F1(C)	Recall(A)	Precision(A)	F1(A)	Recall(S)	Precision(S)	F1(S)
PC	0.80	0.62	0.70	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
ANM	0.20	0.20	0.20	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
CCI	0.20	0.20	0.20	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
HC	0.80	0.73	0.76	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GES	0.70	0.50	0.58	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
GOBNILP	0.60	0.67	0.63	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
SBPN	0.80	0.80	0.80	0.67	1.00	0.80	N.A.	N.A.	N.A.

Table 12

Attributes of the urinary system disease dataset.

Att.	Descriptions	Att.	Descriptions
X_1	Temperature of patient	X_5	Micturition pains
X_2	Occurrence of nausea	X_6	Burning of urethra
X_3	Lumbar pain	X_7	Inflammation of urinary bladder
X_4	Urine pushing	X_8	Nephritis of renal pelvis origin

performance of the proposed SBPN on the datasets with 10 % noise is better than that on the datasets with 5 % noise in terms of capturing synergy effects. The reasons for this are as follows: Noise is a prerequisite for the synergy effects. A high noise proportion implies that the synergy proposition can carry more synergy information regarding the reasoning rule. Thus, the synergy effects are more easily captured.

6.2. Performance in mining “AND/OR” relations and synergy effects on real-world structures

This section focuses on the performance of the SBPN in mining “AND/OR” relations and synergy effects on real-world structures. Note that this study does not employ the classic Bayesian networks⁵ as the benchmark networks, because such networks do not contain any “AND” relation or synergy effect. Instead, the corrosion failure case discussed in [23] (denoted by Case 1) and the aluminum electrolysis cell case presented in [40] (denoted by Case 2) are used to test the performance of SBPN in mining “AND/OR” relations and synergy effects, since they contain ground truth “AND/OR” relations or synergy effects. Fig. 11(a) and Fig. 12(a) present the causal networks for Cases 1 and 2, respectively. As can be seen, Case 1 has six common causal relations, three minimal “AND” units, and five synergy effects. Case 2 has ten common causal relations, three minimal “AND” units, and zero synergy effect. The data generation mechanism for the two cases is the same as that for the synthetic structure. We compare the SBPN against its six peers by fixing the sample size to 1000 and the noise proportion to 0.1 for the two cases.⁶

The mining results are shown in Figs. 11 and 12 and are quantitatively analyzed in Tables 10 and 11, respectively. As can be seen, SBPN performs better than its six peers in capturing “AND/OR” relations and synergy effects in both cases. This result is similar to that discussed in the previous section.

In addition, the urinary system disease dataset [34] and student performance dataset [35] are also used to test the performance of SBPN in mining “AND/OR” relations and synergy effects. They have 120 and 395 cases, respectively, and their attributes are listed in Tables 12 and 13, respectively. Owing to the limited number of “AND” relations and synergy effects in these two datasets, this study only provides a qualitative comparison analysis of all mining models in terms of identifying “AND” relations and synergy effects.

For the urinary system disease dataset, a causal network is constructed based on domain knowledge [17,36], as shown in Fig. 13. It

⁵ <https://www.bnlearn.com/bnrepository>.

⁶ https://github.com/netlearninglabs/SBPN/tree/main/real_world_rule.

Table 13
Attributes of the student performance dataset.

Att.	Descriptions	Att.	Descriptions	Att.	Descriptions
Y_1	Student's school	Y_{12}	Student's guardian	Y_{23}	With a romantic relationship
Y_2	Student's sex	Y_{13}	Home to school travel time	Y_{24}	Quality of family relationships
Y_3	Student's age	Y_{14}	Weekly study time	Y_{25}	Free time after school
Y_4	Student's home address	Y_{15}	Number of past class failures	Y_{26}	Going out with friends
Y_5	Family size	Y_{16}	Extra educational support	Y_{27}	Workday alcohol consumption
Y_6	Parent's cohabitation status	Y_{17}	Family educational support	Y_{28}	Weekend alcohol consumption
Y_7	Mother's education	Y_{18}	Extra paid classes	Y_{29}	Current health status
Y_8	Father's education	Y_{19}	Extra-curricular activities	Y_{30}	Number of school absences
Y_9	Mother's job	Y_{20}	Attended nursery school	Y_{31}	First period grade
Y_{10}	Father's job	Y_{21}	Wanting to obtain higher education	Y_{32}	Second period grade
Y_{11}	Reason to choose this school	Y_{22}	Internet access at home	Y_{33}	Final grade

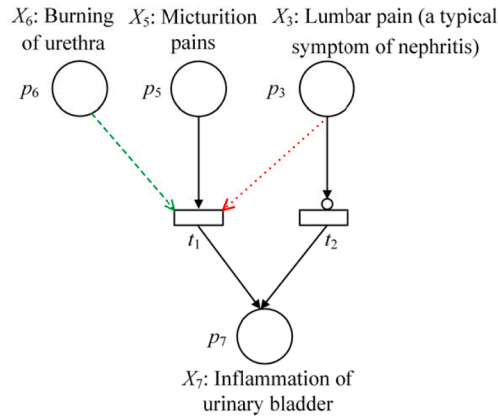


Fig. 13. Causal structure constructed using domain knowledge [17,36].

Table 14
Mining results w.r.t. the two real-world datasets.

Mining result	PC	ANM	CCI	HC	GES	GOBNILP	SBPN
$X_3 \rightarrow X_7$	✓	✓	✓	×	×	×	✓
$X_5 \rightarrow X_7$	✓	×	×	×	✓	×	✓
$X_5 \xrightarrow{X_6(+)} X_7$	—	—	×	—	—	—	✓
$X_5 \xrightarrow{X_3(-)} X_7$	—	—	—	—	—	—	✓
$Y_2 \& Y_{13} \rightarrow Y_{27}$	—	—	—	—	—	—	✓
$Y_{25} \rightarrow Y_{26}$	✓	×	×	×	×	×	✓
$Y_3 \& Y_{15} \rightarrow Y_{21}$	—	—	—	—	—	—	✓
$Y_{25} \xrightarrow{Y_{24}(+)} Y_{26}$	—	—	×	—	—	—	✓
$Y_3 \& Y_{15} \xrightarrow{Y_{29}(-)} Y_{21}$	—	—	—	—	—	—	✓

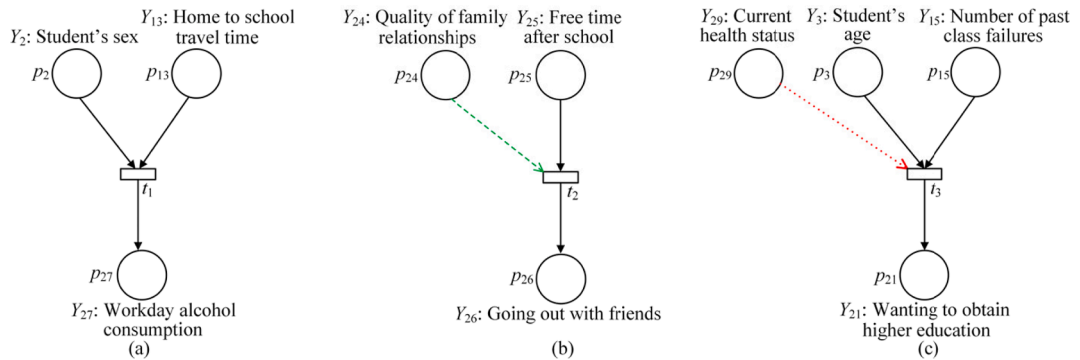


Fig. 14. “AND” relations and synergy effects discovered by SBPN.

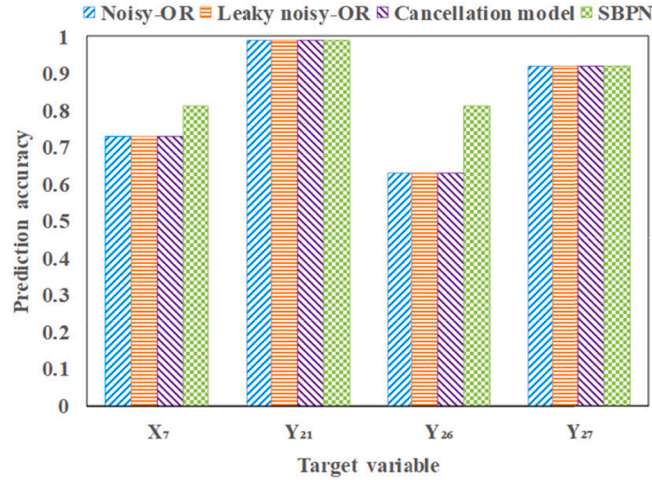


Fig. 15. Prediction performance of different models.

is used to evaluate the knowledge mining performance of all models. In fact, propositions X_7 (inflammation of urinary bladder) and X_8 (nephritis of renal pelvis origin) are negatively correlated in the dataset, and X_3 (lumbar pain) is a typical symptom of X_8 . Therefore, there is a logic “negation” relation between X_3 and X_7 . As shown in Fig. 13, a directed arc terminated with a small circle is employed to represent this relationship.

Additionally, the following conclusions can be drawn from Fig. 13: 1) propositions X_5 and X_3 are the causes of X_7 , whereas X_6 (burning of urethra) is not a cause of X_7 ; 2) X_6 has a catalytic effect on rule R_1 : IF X_5 THEN X_7 . This indicates that the probability of X_7 's occurrence conditioned on X_5 's and X_6 's occurrences is higher than the probability of X_7 's occurrence conditioned on X_5 's occurrence; and 3) X_3 has an inhibitory effect on R_1 . This indicates that the probability of X_7 's occurrence conditioned on X_3 's and X_5 's occurrences is lower than the probability of X_7 's occurrence conditioned on X_5 's occurrence.

The performance of all comparison models in mining the above relations is presented in Table 14, where “✓” indicates that a relation can be identified by a mining algorithm, “×” indicates that a relation cannot be identified by a mining algorithm, and “—” indicates that a mining algorithm does not support mining the type of relation. From Table 14, the PC and SBPN can identify the antecedent-consequence relation from X_3 or X_5 to X_7 . More importantly, the SBPN is capable of capturing the catalytic effect of X_6 on R_1 (denoted by $X_5 \xrightarrow{X_6(+)} X_7$). Although CCI supports the mining of such a catalytic effect, it fails to discover this effect. The PC, ANM, HC, GES, and GOBNILP are unable to deal with this effect, since they can only discover common causal relations. In addition, SBPN can capture the inhibitory effect of X_3 on R_1 (denoted by $X_5 \xrightarrow{X_3(-)} X_7$). However, its six peers cannot deal with this effect as they do not consider the weakening effect between the causal variables X_5 and X_3 .

Some meaningful reasoning rules discovered by the SBPN on the student performance dataset are shown in Fig. 14. As can be seen from Fig. 14: 1) Y_{27} (workday alcohol consumption) is jointly determined by Y_2 (student's sex) and Y_{13} (home to school travel time); 2) Y_{25} (free time after school) determines Y_{26} (going out with friends). Although Y_{24} (quality of family relationship) is not a decisive factor of Y_{26} , the better the quality of family relationships, the higher the probability of going out with friends; and 3) Y_{29} (current health status) is not a determining factor of Y_{21} (wanting to obtain higher education). However, on the premise that Y_3 (student's age) and Y_{15} (number of past class failures) are known, the worse the health status, the lower the possibility of wanting to obtain higher education. From Table 14, SBPN's six peers fail to discover the above “AND” relation and synergy effects.

6.3. Probabilistic reasoning performance in the presence of synergy effects

Without considering synergy effects, an SBPN can be considered as a leaky noisy-OR model [15]. In this case, the reasoning performance of the SBPN is similar to that of the leaky noisy-OR model. Therefore, this study focuses only on the probabilistic reasoning performance of the SBPN in the presence of synergy effects. Variable X_7 in Fig. 13 and variables Y_{21} , Y_{26} , and Y_{27} in Fig. 14 are selected as prediction variables because most of these variables are affected by synergy effects. The PN-based knowledge reasoning models [21–23] are not selected as its peers because their model parameters are mainly given by experts. The noisy-OR model [13], leaky noisy-OR model [15], and cancellation model [18] are employed for comparison because the reasoning parameters of these models can be learned from the data. More importantly, their reasoning mechanism involves the concept of synergy effects. Specifically, the noisy-OR model assumes that causes interact with each other in the form of logic “OR”. The leaky noisy-OR model (an improved version of the noisy-OR model) considers the impact of hidden variables on probabilistic reasoning. The cancellation model can depict the weakening effects among causes.

Fig. 15 shows the prediction accuracy of the four reasoning models for all target variables. Clearly, the SBPN performs better than its three peers. The reasons for this are as follows: A cancellation model focuses on the weakening effects among the causal variables,

whereas a noisy-OR model and a leaky noisy-OR model can only deal with the enhancement effects among the causal variables. None of the studies consider the interactions between causal and non-causal variables. The SBPN can effectively handle the enhancement and weakening effects among causal variables, as well as among causal and non-causal variables. This makes the probabilistic reasoning more robust.

7. Conclusions

In this research, SBPN is proposed to characterize “AND/OR” relations and synergy effects during knowledge mining, representation, and reasoning. In SBPN, PN transitions are the core of knowledge modelling, providing a new paradigm for representing and reasoning of complex causal structures with synergy effects. Specifically, this study first presents a formal definition of SBPN by integrating the conditional probability distribution of BN with the structure of PN, and by defining the synergy effects on PN transitions. Then, the modelling paradigm for “AND/OR” relations and synergy effects is presented based on the PN structure. On this basis, the following three steps are taken to mine “AND/OR” relations and synergy effects from data: 1) determine the antecedent-consequence relations among propositions; 2) identify the “AND/OR” relations among antecedent propositions; and 3) capture the synergy effects of propositions on rules. Furthermore, to quantify the impact of synergy effects on probabilistic reasoning, this study introduces synergy effects and transitions into a leaky noisy-OR model, and proposes the idea of learning synergy coefficients from data. The experimental results well demonstrate the effectiveness of the proposed SBPN in mining “AND/OR” relations and synergy effects, as well as its advantage in probabilistic reasoning in the presence of synergy effects.

However, for “AND” structures and “OR” structures pointing to the same effect variable, identifying the “AND” structures becomes a challenging issue as the number of “OR” structures increases. This is because the greater the number of “OR” structures, the greater their impact on the value of the effect variable. In this case, it is difficult to identify the “AND” relations among causal variables. This issue is worth studying in the future. Another focus in future works is to discover the “AND/OR” relations and synergy effects from temporal data.

CRedit authorship contribution statement

Xiaoliang Wang: Writing – review & editing, Writing – original draft, Methodology. **Faming Lu:** Supervision, Conceptualization. **MengChu Zhou:** Writing – review & editing. **Qingtian Zeng:** Funding acquisition. **Yunxia Bao:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ins.2024.121019>.

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